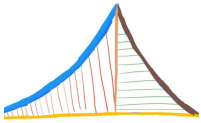
Math for EE



G. V. V. Sharma

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

License: https://creativecommons.org/licenses/by-sa/3.0/
and
https://www.gnu.org/licenses/fdl-1.3.en.html

First manual appeared in January 2018
First edition published on July 10, 2024

In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

CONTENTS

1 CBSE 4
1.1 Formulae-12th 4
1.2 Problems-12th 7
1.3 Formulae-10th 62
1.4 Problems-10th 65

1. CBSE

A. Formulae-12th

1.1.1

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a\left(ei - fh\right) - b\left(di - fg\right) + c\left(dh - eg\right) \tag{1.1.1.1}$$

1.1.2 King's Rule

$$\int_a^b f\left(x\right)\, dx = \int_a^b f\left(a + b - x\right)\, dx \tag{1.1.2.1}$$

For the interval $[0, a]$,

$$\int_0^a f\left(x\right)\, dx = \int_0^a f\left(a - x\right)\, dx \tag{1.1.2.2}$$

Hence,

$$2\int_0^a f\left(x\right)\, dx = \int_0^a \left[f\left(x\right) + f\left(a - x\right)\right]\, dx \tag{1.1.2.3}$$

1.1.3 Integration by Parts

$$\int u\, dv = uv - \int v\, du \tag{1.1.3.1}$$

For definite integrals,

$$\int_a^b u\, dv = \left[uv\right]_a^b - \int_a^b v\, du \tag{1.1.3.2}$$

1.1.4 Integration by Substitution

$$u = g\left(x\right), \qquad du = g'\left(x\right)\, dx \tag{1.1.4.1}$$

$$\int f\left(g\left(x\right)\right) g'\left(x\right)\, dx = \int f\left(u\right)\, du \tag{1.1.4.2}$$

For definite integrals,

$$\int_a^b f\left(g\left(x\right)\right) g'\left(x\right)\, dx = \int_{g(a)}^{g(b)} f\left(u\right)\, du \tag{1.1.4.3}$$

$$\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + C \tag{1.1.4.4}$$

1.1.5 Partial Fractions

For distinct constants a and b,

$$\frac{1}{(x-a)(x-b)} = \frac{1}{a-b}\left[\frac{1}{x-a} - \frac{1}{x-b}\right] \tag{1.1.5.1}$$

$$\int \frac{dx}{(x-a)(x-b)} = \frac{1}{a-b}\ln\left|\frac{x-a}{x-b}\right| + C \tag{1.1.5.2}$$

$$\frac{1}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b} \tag{1.1.5.3}$$

where

$$A = \frac{1}{a-b}, \qquad B = -\frac{1}{a-b} \tag{1.1.5.4}$$

1.1.6 Finding Area

Area under the curve $y = f(x)$, where $f(x) \geq 0$,

$$A = \int_a^b f(x)\,dx \tag{1.1.6.1}$$

Area between a curve and the x-axis,

$$A = \int_a^b |f(x)|\,dx \tag{1.1.6.2}$$

If the curve crosses the x-axis, split the integral at its points of intersection.
Area between two curves, where $f(x) \geq g(x)$,

$$A = \int_a^b [f(x) - g(x)]\,dx \tag{1.1.6.3}$$

In general,

$$A = \int_a^b |f(x) - g(x)|\,dx \tag{1.1.6.4}$$

Area with respect to y, for the curve $x = f(y)$ and the y-axis,

$$A = \int_c^d |f(y)|\, dy \tag{1.1.6.5}$$

Area between two curves with respect to y,

$$A = \int_c^d [x_{\text{right}}(y) - x_{\text{left}}(y)]\, dy \tag{1.1.6.6}$$

1.1.7 Let $\mathbf{a}$ and $\mathbf{b}$ are two vectors then

$$\mathbf{a}.\mathbf{b} = |a||b|cos\theta \tag{1.1.7.1}$$

where θ is the angle between two vectors.

1.1.8

$$\frac{d}{dx}x^n = nx^{n-1} \tag{1.1.8.1}$$

1.1.9 The conditional probability of event S given event E is defined as

$$P(S \mid E) = \frac{P(S \cap E)}{P(E)}. \tag{1.1.9.1}$$

1.1.10

$$\frac{d}{dx}\left(\tan^{-1} u\right) = \frac{1}{1+u^2}\frac{du}{dx}. \tag{1.1.10.1}$$

1.1.11 Let $\mathbf{A} = x_1\mathbf{i} + y_1\mathbf{j} + z_1\mathbf{k}$ and $\mathbf{B} = x_2\mathbf{i} + y_2\mathbf{j} + z_2\mathbf{k}$ be two vectors then the vector perpendicular to both the vectors is

$$\mathbf{A} \times \mathbf{B} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix} \tag{1.1.11.1}$$

1.1.12 Let $\mathbf{A}$ be a vector then the unit vector in the direction of vector $\mathbf{A}$ is $\dfrac{\mathbf{A}}{|A|}$

1.1.13 $\dfrac{de^{ax}}{dx} = e^{ax}a$

1.1.14 Linear first-order differential equation
For

$$\frac{dy}{dx} + P(x)y = Q(x) \tag{1.1.14.1}$$

The integrating factor is

$$\text{I.F.} = e^{\int P(x)\, dx} \tag{1.1.14.2}$$

The solution is

$$y\, e^{\int P(x)\, dx} = \int Q(x) e^{\int P(x)\, dx}\, dx + C \tag{1.1.14.3}$$

1.1.15 $\frac{d}{dx} sin^{-1}x = \frac{1}{\sqrt{1-x^2}}$

1.1.16 Bayes' theorem ,

$$P(E \mid A) = \frac{P(E)\,P(A \mid E)}{P(A)}. \tag{1.1.16.1}$$

1.1.17 The following Laplace transform formulae are used in the solutions.

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\,dt, \tag{1.1.17.1}$$

$$\mathcal{L}\left\{\frac{df}{dt}\right\} = sF(s) - f(0), \tag{1.1.17.2}$$

$$\mathcal{L}\{tf(t)\} = -\frac{dF(s)}{ds}, \tag{1.1.17.3}$$

$$\mathcal{L}\{1\} = \frac{1}{s}, \tag{1.1.17.4}$$

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, \qquad n > -1, \tag{1.1.17.5}$$

$$E_1(s) = \int_s^\infty \frac{e^{-u}}{u}\,du, \tag{1.1.17.6}$$

$$\mathcal{L}\left\{\frac{1}{1+t}\right\} = e^s E_1(s), \tag{1.1.17.7}$$

$$\mathcal{L}\{\ln(1+t)\} = \frac{e^s E_1(s)}{s}, \tag{1.1.17.8}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{n!}. \tag{1.1.17.9}$$

B. Problems-12th

1.2.1 Let A be a scalar matrix of order 3 such that the sum of all its elements is 9. Find the product of all the elements of A.

a) 9 b) 27 c) 3 d) 0

Solution: Since A is a scalar matrix of order 3, it can be written as

$$A = \begin{pmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{pmatrix} \tag{1.2.1.1}$$

Given that the sum of all the elements is 9,

$$a + a + a = 9 \tag{1.2.1.2}$$

$$3a = 9 \tag{1.2.1.3}$$

$$a = 3 \tag{1.2.1.4}$$

Therefore, the matrix is

$$A = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} \tag{1.2.1.5}$$

The product of all the elements is

$$3 \times 0 \times 0 \times 0 \times 3 \times 0 \times 0 \times 0 \times 3 = 0 \tag{1.2.1.6}$$

Hence, the correct answer is

$$\text{Option } (D)\text{: } 0 \tag{1.2.1.7}$$

1.2.2 Let $f : [0, \infty) \to [-5, \infty)$ be defined by

$$f(x) = 9x^2 + 6x - 5. \tag{1.2.2.1}$$

Then f is

a) one-one but not onto
b) onto but not one-one
c) both one-one and onto
d) neither one-one nor onto

Solution: Given,

$$f(x) = 9x^2 + 6x - 5, \qquad x = 0. \tag{1.2.2.2}$$

Step 1: Check whether the function is one-one.
Differentiating with respect to x,

$$f'(x) = 18x + 6. \tag{1.2.2.3}$$

Since $x = 0$,

$$18x + 6 > 0. \tag{1.2.2.4}$$

Therefore, $f(x)$ is strictly increasing on $[0, \infty)$.
Hence, f is a one-one function.
Step 2: Check whether the function is onto.
At $x = 0$,

$$f(0) = 9(0)^2 + 6(0) - 5 = -5. \tag{1.2.2.5}$$

Also,

$$\lim_{x \to \infty} f(x) = \infty. \tag{1.2.2.6}$$

Since f is continuous and strictly increasing on $[0, \infty)$, its range is

$$\text{Range}(f) = [-5, \infty). \tag{1.2.2.7}$$

The range of f is equal to its codomain. Therefore, f is onto.
Conclusion:

The function is both one-one and onto.
Hence, the correct answer is

$$\text{Option } (C)\text{: Both one-one and onto} \tag{1.2.2.8}$$

1.2.3 If

$$\begin{vmatrix} -a & a & a \\ b & -b & b \\ c & c & -c \end{vmatrix} = kabc, \tag{1.2.3.1}$$

then the value of k is

a) 0 b) 1 c) 2 d) 4

Solution: Given,

$$\begin{vmatrix} -a & a & a \\ b & -b & b \\ c & c & -c \end{vmatrix} = kabc, \tag{1.2.3.2}$$

Taking a, b, and c common from the first, second, and third rows respectively, we get

$$= abc \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} \tag{1.2.3.3}$$

using the (1.1.1.1) expanding the determinant.

$$= abc\left[(-1)(1-1) - (-1-1) + (1+1)\right]. = abc[0+2+2] \quad = 4abc. \tag{1.2.3.4}$$

Comparing this result with $kabc$, we obtain

$$kabc = 4abc. \tag{1.2.3.5}$$

Therefore,

$$k = 4 \tag{1.2.3.6}$$

Hence, the correct answer is

1.2.4 The number of points at which the function

$$f(x) = \begin{cases} |x| + 3, & x \le -3, \\ -2x, & -3 < x < 3, \\ 6x + 2, & x = 3 \end{cases}$$

is discontinuous is

a) 0 b) 1 c) 2 d) 3

Solution:
The function is defined by three different expressions. We check continuity at the points where the definition changes: $x = -3$ and $x = 3$.
Step 1: Check continuity at $x = -3$.
The left-hand limit is

$$\lim_{x \to -3^-} f(x) = |-3| + 3 = 6. \tag{1.2.4.1}$$

The right-hand limit is

$$\lim_{x \to -3^+} f(x) = -2(-3) = 6. \tag{1.2.4.2}$$

The function value is

$$f(-3) = |-3| + 3 = 6. \tag{1.2.4.3}$$

Therefore,

$$\lim_{x \to -3^-} f(x) = \lim_{x \to -3^+} f(x) = f(-3). \tag{1.2.4.4}$$

Hence, f is continuous at $x = -3$.
Step 2: Check continuity at $x = 3$.
The left-hand limit is

$$\lim_{x \to 3^-} f(x) = -2(3) = -6. \tag{1.2.4.5}$$

The right-hand limit is

$$\lim_{x \to 3^+} f(x) = 6(3) + 2 = 20. \tag{1.2.4.6}$$

The function value is

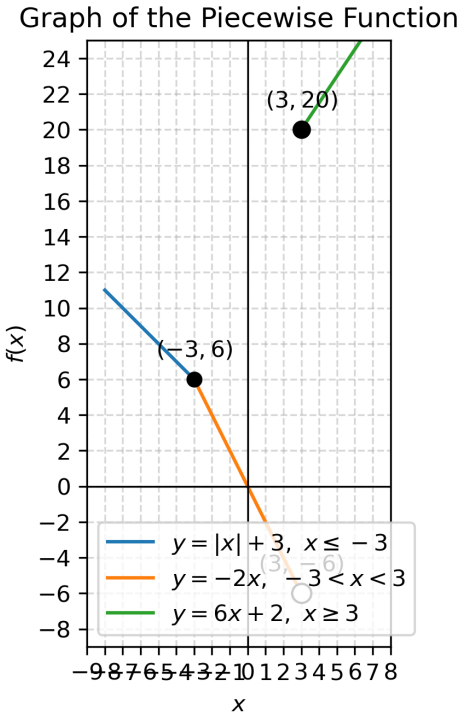
$$f(3) = 6(3) + 2 = 20. \tag{1.2.4.7}$$

Since

$$\lim_{x \to 3^-} f(x) \neq \lim_{x \to 3^+} f(x), \tag{1.2.4.8}$$

the function is discontinuous at $x = 3$.
Conclusion:
The function has exactly one point of discontinuity.
The following code generates the figure

codes/s124.py



1.2.5 The function

$$f\left(x\right)=x^{3}-3x^{2}+12x-18 \tag{1.2.5.1}$$

is

a) strictly decreasing on $\mathbb{R}$
b) strictly increasing on $\mathbb{R}$
c) neither increasing nor decreasing on $\mathbb{R}$
d) a constant function on $\mathbb{R}$

Solution: Given,

$$f\left(x\right)=x^{3}-3x^{2}+12x-18. \tag{1.2.5.2}$$

Differentiating with respect to x, we obtain

$$f'\left(x\right)=3x^{2}-6x+12. \tag{1.2.5.3}$$

Taking 3 common,

$$f'\left(x\right)=3\left(x^{2}-2x+4\right). \tag{1.2.5.4}$$

Completing the square,

$$f'(x) = 3\left[(x-1)^2 + 3\right]. \tag{1.2.5.5}$$

Since

$$(x-1)^2 = 0, \tag{1.2.5.6}$$

we have

$$(x-1)^2 + 3 > 0. \tag{1.2.5.7}$$

Therefore,

$$f'(x) > 0 \qquad \text{for all } x \in \mathbb{R}. \tag{1.2.5.8}$$

Since the derivative is positive throughout $\mathbb{R}$, the function is strictly increasing on $\mathbb{R}$.

1.2.6 Evaluate

$$I = \int_0^{\pi/2} \frac{\sin x - \cos x}{1 + \sin x \cos x}\, dx. \tag{1.2.6.1}$$

(A) 1 (B) 0 (C) -1 (D) $\frac{\pi}{2}$

Solution: Given,

$$I = \int_0^{\pi/2} \frac{\sin x - \cos x}{1 + \sin x \cos x}\, dx. \tag{1.2.6.2}$$

Using the substitution

$$x = \frac{\pi}{2} - t, \tag{1.2.6.3}$$

we have

$$dx = -dt. \tag{1.2.6.4}$$

When $x = 0$, $t = \frac{\pi}{2}$.
When $x = \frac{\pi}{2}$, $t = 0$.
Also,

$$\sin x = \cos t, \qquad \cos x = \sin t. \tag{1.2.6.5}$$

Therefore,

$$I = \int_0^{\pi/2} \frac{\cos t - \sin t}{1 + \sin t \cos t}\, dt. \tag{1.2.6.6}$$

Rewriting the variable t as x,

$$I = \int_0^{\pi/2} \frac{\cos x - \sin x}{1 + \sin x \cos x}\, dx. \tag{1.2.6.7}$$

Comparing this with the original integral,

$$I = -I. \tag{1.2.6.8}$$

Hence,

$$2I = 0 \tag{1.2.6.9}$$

Therefore,

$$I = 0. \tag{1.2.6.10}$$

1.2.7 Which of the following differential equations is a homogeneous differential equation?

a) $\frac{dy}{dx} = \cos x - \sin\left(\frac{x}{y}\right)$

b) $\frac{dy}{dx} = \frac{x+y}{x-y}$

c) $\frac{dy}{dx} = x + y$

d) $\frac{dy}{dx} = x^2 + y^2$

Solution: A first-order differential equation of the form

$$\frac{dy}{dx} = F(x, y) \tag{1.2.7.1}$$

is homogeneous if $F(x, y)$ can be expressed as a function of $\frac{y}{x}$ alone. For option (B),

$$\frac{dy}{dx} = \frac{x+y}{x-y}. \tag{1.2.7.2}$$

Dividing the numerator and denominator by x, we get

$$\frac{dy}{dx} = \frac{1 + \frac{y}{x}}{1 - \frac{y}{x}}. \tag{1.2.7.3}$$

The right-hand side is a function of $\frac{y}{x}$ alone.

Therefore, the differential equation in option (B) is homogeneous.

1.2.8 For any two vectors $\mathbf{a}$ and $\mathbf{b}$, which of the following statements is always true?

a) $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| + |\mathbf{b}|$

b) $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}|$

c) $\mathbf{a} \cdot \mathbf{b} \leq |\mathbf{a}|\,|\mathbf{b}|$

d) $\mathbf{a} \times \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}|$

Solution: The dot product of two vectors is given by(1.1.7.1)

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}|\,|\mathbf{b}| \cos\theta, \tag{1.2.8.1}$$

where θ is the angle between the vectors. Since

$$-1 \le \cos\theta \le 1, \tag{1.2.8.2}$$

we obtain

$$-|\mathbf{a}|\,|\mathbf{b}| \le \mathbf{a}\cdot\mathbf{b} \le |\mathbf{a}|\,|\mathbf{b}|. \tag{1.2.8.3}$$

Therefore,

$$\mathbf{a}\cdot\mathbf{b} \le |\mathbf{a}|\,|\mathbf{b}|. \tag{1.2.8.4}$$

1.2.9 The foot of the perpendicular drawn from the point $P\,(0,1,2)$ to the x-axis is

a) $(0,1,0)$ b) $(0,0,2)$ c) $(1,0,0)$ d) $(0,0,0)$

Solution: Let the given point be

$$P = (0,1,2)\,. \tag{1.2.9.1}$$

Any point on the x-axis has coordinates

$$Q = (x,0,0)\,. \tag{1.2.9.2}$$

The direction vector of the x-axis is

$$\mathbf{d} = (1,0,0)\,. \tag{1.2.9.3}$$

Since PQ is perpendicular to the x-axis,

$$\overrightarrow{PQ}\cdot\mathbf{d} = 0. \tag{1.2.9.4}$$

Now,

$$\overrightarrow{PQ} = Q - P = (x,-1,-2)\,. \tag{1.2.9.5}$$

Therefore,

$$(x,-1,-2)\cdot(1,0,0) = 0. \tag{1.2.9.6}$$

Expanding the dot product,

$$x\,(1) + (-1)\,(0) + (-2)\,(0) = 0. \tag{1.2.9.7}$$

Hence,

$$x = 0. \tag{1.2.9.8}$$

Therefore, the foot of the perpendicular is

$$Q = (0,0,0)\,. \tag{1.2.9.9}$$

1.2.10 The common region determined by all the constraints in a linear programming problem is called the

a) Objective function

b) Optimal solution
c) Optimal value
d) Feasible region

Solution: In a linear programming problem, the constraints are linear inequalities or equations involving the decision variables. The region containing all points that satisfy every constraint, including the non-negativity restrictions, is called the *feasible region*.
Therefore, the correct answer is

$$\text{Option } (D)\text{: Feasible region} \tag{1.2.10.1}$$

1.2.11 If E and S are two events such that $E \subseteq S$ and $P(E) > 0$, then $P(S \mid E)$ is equal to

a) $P(S)$ b) $P(E)$ c) 1 d) 0

Solution: The conditional probability of event S given event E is defined as

$$P(S \mid E) = \frac{P(S \cap E)}{P(E)}. \tag{1.2.11.1}$$

Since $E \subseteq S$, we have

$$S \cap E = E. \tag{1.2.11.2}$$

Therefore,

$$P(S \mid E) = \frac{P(E)}{P(E)}. \tag{1.2.11.3}$$

Given that $P(E) > 0$, we obtain

$$P(S \mid E) = 1. \tag{1.2.11.4}$$

1.2.12 If $A = [a_{ij}]$ is a 3×3 matrix, where

$$a_{ij} = i - 3j, \tag{1.2.12.1}$$

then which of the following is false?

(A) $a_{11} < 0$ (B) $a_{12} + a_{21} = -6 (C)$ $a_{13} > a_{31}$ (D) $a_{31} = 0$

Solution: Given,

$$a_{ij} = i - 3j \tag{1.2.12.2}$$

Using this formula, we calculate the required matrix elements.
Step 1: Check option (A) For $i = 1$ and $j = 1$,

$$a_{11} = 1 - 3(1) = -2 \tag{1.2.12.3}$$

Since $-2 < 0$, option (A) is true.
Step 2: Check option (B) For $i = 1$ and $j = 2$,

$$a_{12} = 1 - 3\,(2) = -5 \tag{1.2.12.4}$$

For $i = 2$ and $j = 1$,

$$a_{21} = 2 - 3\,(1) = -1 \tag{1.2.12.5}$$

Therefore,

$$a_{12} + a_{21} = -5 + (-1) = -6 \tag{1.2.12.6}$$

Hence, option (B) is true.
Step 3: Check option (C) For $i = 1$ and $j = 3$,

$$a_{13} = 1 - 3\,(3) = -8 \tag{1.2.12.7}$$

For $i = 3$ and $j = 1$,

$$a_{31} = 3 - 3\,(1) = 0 \tag{1.2.12.8}$$

Option (C) states that

$$a_{13} > a_{31} \tag{1.2.12.9}$$

Substituting the values,

$$-8 > 0 \tag{1.2.12.10}$$

This statement is false.
Step 4: Check option (D) We have already calculated

$$a_{31} = 0 \tag{1.2.12.11}$$

Therefore, option (D) is true.

1.2.13 The derivative of $\tan^{-1}\left(x^2\right)$ with respect to x is:

a) $\dfrac{x}{1+x^4}$ b) $\dfrac{2x}{1+x^4}$ c) $\dfrac{-2x}{1+x^4}$ d) $\dfrac{1}{1+x^4}$

Solution: Given,

$$y = \tan^{-1}\left(x^2\right) \tag{1.2.13.1}$$

from (**??**) the derivative of $tan^{-1}u$ is

$$\frac{d}{dx}\left(\tan^{-1} u\right) = \frac{1}{1+u^2}\frac{du}{dx}. \tag{1.2.13.2}$$

Here, $u = x^2$, so

$$\frac{du}{dx} = 2x. \tag{1.2.13.3}$$

Applying the formula(1.2.13.2) on (1.2.13.1),

$$\frac{dy}{dx} = \frac{1}{1+\left(x^2\right)^2} \cdot 2x. \tag{1.2.13.4}$$

Simplifying,

$$\frac{dy}{dx} = \frac{2x}{1+x^4}. \tag{1.2.13.5}$$

1.2.14 The degree of the differential equation

$$\left(y''\right)^2 + \left(y'\right)^3 = x\sin\left(y'\right) \tag{1.2.14.1}$$

is:

a) 1 b) 2 c) 3 d) Not defined

Solution: The degree of a differential equation is the highest power of the highest-order derivative, provided the equation is a polynomial in its derivatives. The given equation is

$$\left(y''\right)^2 + \left(y'\right)^3 = x\sin\left(y'\right). \tag{1.2.14.2}$$

Rearranging,

$$\left(y''\right)^2 + \left(y'\right)^3 - x\sin\left(y'\right) = 0. \tag{1.2.14.3}$$

The equation contains the term

$$\sin\left(y'\right), \tag{1.2.14.4}$$

which is a trigonometric function of the derivative y'. Therefore, the equation is not a polynomial in its derivatives. Hence, its degree is not defined.

1.2.15 The unit vector perpendicular to both vectors $\mathbf{i}+\mathbf{k}$ and $\mathbf{i}-\mathbf{k}$ is:

a) $2\mathbf{j}$ b) $\mathbf{j}$ c) $\frac{\mathbf{i}-\mathbf{k}}{\sqrt{2}}$ d) $\frac{\mathbf{i}+\mathbf{k}}{\sqrt{2}}$

Solution: Let

$$\mathbf{A} = \mathbf{i}+\mathbf{k} \tag{1.2.15.1}$$

and

$$\mathbf{B} = \mathbf{i}-\mathbf{k}. \tag{1.2.15.2}$$

A vector perpendicular to both $\mathbf{A}$ and $\mathbf{B}$ can be obtained using their cross product(1.1.11.1). Step 1: Calculate the cross product

$$\mathbf{A}\times\mathbf{B} = \begin{pmatrix} 0-0 \\ 1-(-1) \\ 0-0 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} \tag{1.2.15.3}$$

Step 2: Calculate the magnitude

$$|\mathbf{A}\times\mathbf{B}| = 2. \tag{1.2.15.4}$$

Step 3: Finding the unit vector
The unit vector is obtained by dividing the vector by its magnitude(1.1.12).

$$\mathbf{n} = \frac{\mathbf{A}\times\mathbf{B}}{|\mathbf{A}\times\mathbf{B}|} \tag{1.2.15.5}$$

$$= \frac{2\mathbf{j}}{2} \tag{1.2.15.6}$$

$$= \mathbf{j}. \tag{1.2.15.7}$$

1.2.16 Direction ratios of a vector parallel to the line

$$\frac{2x-1}{2} = -y = \frac{6z+1}{3} \tag{1.2.16.1}$$

are:

a) $2,-1,6$ b) $2,1,6$ c) $2,1,3$ d) $2,-1,3$

Solution: The symmetric equation of a line is

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}, \tag{1.2.16.2}$$

where a,b,c are the direction ratios of the line.
The given equation is

$$\frac{2x-1}{2} = -y = \frac{6z+1}{3}. \tag{1.2.16.3}$$

Rewrite each expression in standard form:

$$\frac{x-\frac{1}{2}}{1} = \frac{y-0}{-\frac{1}{2}} = \frac{z+\frac{1}{6}}{\frac{1}{2}}. \tag{1.2.16.4}$$

Therefore, direction ratios are

$$1,-\frac{1}{2},\frac{1}{2}. \tag{1.2.16.5}$$

Multiplying each by 2 gives the proportional direction ratios

$$2,-1,1. \tag{1.2.16.6}$$

1.2.17 If

$$F(x) = [F(x)]^2 = \begin{pmatrix} \cos x & \sin x & 0 \\ -\sin x & \cos x & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{1.2.17.1}$$

and

$$[F(x)]^2 = F(kx), \tag{1.2.17.2}$$

then the value of k is:

$(A)\ 1 \qquad (B)\ 2 \qquad (C)\ 0 \qquad (D)\ -2$

Solution:
Given,

$$F(x) = [F(x)]^2 = \begin{pmatrix} \cos x & \sin x & 0 \\ -\sin x & \cos x & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{1.2.17.3}$$

We need to find k such that

$$[F(x)]^2 = F(kx). \tag{1.2.17.4}$$

Step 1: Calculate $[F(x)]^2$
Multiplying $F(x)$ by itself, we get

$$[F(x)]^2 = \begin{pmatrix} \cos^2 x - \sin^2 x & 2\sin x \cos x & 0 \\ -2\sin x \cos x & \cos^2 x - \sin^2 x & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{1.2.17.5}$$

Using the double-angle identities,

$$\cos^2 x - \sin^2 x = \cos 2x \tag{1.2.17.6}$$

and

$$2\sin x \cos x = \sin 2x, \tag{1.2.17.7}$$

we obtain

$$[F(x)]^2 = \begin{pmatrix} \cos 2x & \sin 2x & 0 \\ -\sin 2x & \cos 2x & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{1.2.17.8}$$

Step 2: Compare with $F(kx)$
By definition,

$$F(kx) = \begin{pmatrix} \cos kx & \sin kx & 0 \\ -\sin kx & \cos kx & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{1.2.17.9}$$

Comparing the matrices gives

$$\cos kx = \cos 2x, \qquad \sin kx = \sin 2x. \tag{1.2.17.10}$$

Therefore,

$$kx = 2x, \tag{1.2.17.11}$$

so

$$k = 2. \tag{1.2.17.12}$$

1.2.18 If a line makes an angle of 30° with the positive direction of the x-axis and 120° with the positive direction of the y-axis, then the angle which it makes with the positive direction of the z-axis is:

$(A)\ 90^\circ$ $(B)\ 120^\circ$ $(C)\ 60^\circ$ $(D)\ 0^\circ$

Solution: Let α, β, and γ be the angles made by the line with the positive directions of the x-, y-, and z-axes, respectively.
The direction cosines satisfy the identity

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1. \tag{1.2.18.1}$$

Given,

$$\alpha = 30^\circ, \qquad \beta = 120^\circ. \tag{1.2.18.2}$$

Substituting α and β in (1.2.18.1),

$$\cos^2 30^\circ + \cos^2 120^\circ + \cos^2\gamma = 1. \tag{1.2.18.3}$$

Using

$$\cos 30^\circ = \frac{\sqrt{3}}{2}, \qquad \cos 120^\circ = -\frac{1}{2}, \tag{1.2.18.4}$$

we get

$$\frac{3}{4} + \frac{1}{4} + \cos^2\gamma = 1. \tag{1.2.18.5}$$

Therefore,

$$1 + \cos^2\gamma = 1, \tag{1.2.18.6}$$

so

$$\cos^2\gamma = 0. \tag{1.2.18.7}$$

Hence,

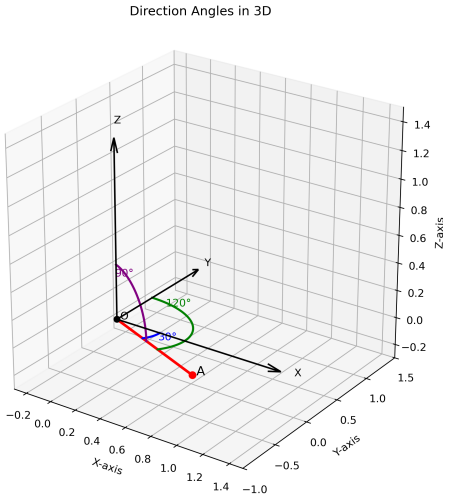
$$\cos\gamma = 0. \tag{1.2.18.8}$$

Since $0^\circ \le \gamma \le 180^\circ$,

$$\gamma = 90^\circ. \tag{1.2.18.9}$$

The following code generates the figure

codes/s1218.py



1.2.19 ASSERTION : For any symmetric matrix A, $B'AB$ is a skew-symmetric matrix.
REASON : A square matrix P is skew-symmetric if $P' = -P$.
Choose the correct option:

a) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
b) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion .
c) Assertion is true, but Reason is false.
d) Assertion is false, but Reason is true.

Solution: Step 1: Check Assertion Given that A is a symmetric matrix,

$$A' = A. \tag{1.2.19.1}$$

Let

$$P = B'AB. \tag{1.2.19.2}$$

Taking the transpose of P, we get

$$P' = (B'AB)'. \tag{1.2.19.3}$$

Using the transpose property $(XYZ)' = Z'Y'X'$,

$$P' = B'A'B. \tag{1.2.19.4}$$

Since $A' = A$,

$$P' = B'AB. \tag{1.2.19.5}$$

Therefore,

$$P' = P. \tag{1.2.19.6}$$

Hence, $B'AB$ is symmetric, not necessarily skew-symmetric. Thus, Assertion is false.
Step 2: Check Reason. A square matrix P is skew-symmetric if

$$P' = -P. \tag{1.2.19.7}$$

This is the correct definition of a skew-symmetric matrix. Therefore, Reason is true.
Assertion is false, but Reason is true.

1.2.20 ASSERTION : For two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$,

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}. \tag{1.2.20.1}$$

REASON : For two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$,

$$\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{a}. \tag{1.2.20.2}$$

Choose the correct option:

a) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion .
b) Both Assertion and Reason are true, but Reason (R) is not the correct explanation of Assertion .
c) Assertion is true, but Reason is false.
d) Assertion is false, but Reason is true.

Solution: Step 1: Verify Assertion
The dot product of two vectors is commutative.
Using the formula (1.1.1.1)

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta, \tag{1.2.20.3}$$

where θ is the angle between the vectors, we get

$$\mathbf{b} \cdot \mathbf{a} = |\mathbf{b}||\mathbf{a}| \cos\theta. \tag{1.2.20.4}$$

Since multiplication of real numbers is commutative,

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}. \tag{1.2.20.5}$$

Therefore, Assertion is true.
Step 2: Verify Reason
The cross product of two vectors is anti-commutative:

$$\mathbf{a} \times \mathbf{b} = -\left(\mathbf{b} \times \mathbf{a}\right). \tag{1.2.20.6}$$

In general, therefore,

$$\mathbf{a} \times \mathbf{b} \neq \mathbf{b} \times \mathbf{a}. \tag{1.2.20.7}$$

Hence,Reason is false.
Assertion is true, but Reason is false.

1.2.21 Find the value of

$$\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)+\cot^{-1}\left(\frac{1}{\sqrt{3}}\right)+\tan^{-1}\left[\sin\left(-\frac{\pi}{2}\right)\right]. \tag{1.2.21.1}$$

Solution: We know that

$$\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)=-\frac{\pi}{6}. \tag{1.2.21.2}$$

Also, using the principal value of the inverse cotangent,

$$\cot^{-1}\left(\frac{1}{\sqrt{3}}\right)=\frac{\pi}{3}. \tag{1.2.21.3}$$

Since

$$\sin\left(-\frac{\pi}{2}\right)=-1, \tag{1.2.21.4}$$

we have

$$\tan^{-1}\left[\sin\left(-\frac{\pi}{2}\right)\right]=\tan^{-1}(-1)=-\frac{\pi}{4}. \tag{1.2.21.5}$$

Substituting these values in (1.2.21.1), the given expression becomes

$$=-\frac{\pi}{6}+\frac{\pi}{3}-\frac{\pi}{4}. \tag{1.2.21.6}$$

Taking the LCM of 6, 3, and 4, we obtain

$$-\frac{2\pi}{12}+\frac{4\pi}{12}-\frac{3\pi}{12} \tag{1.2.21.7}$$

$$=\frac{(-2+4-3)\,\pi}{12} \tag{1.2.21.8}$$

$$=-\frac{\pi}{12}. \tag{1.2.21.9}$$

1.2.22 Find the domain and range of the function

$$f(x)=\sin^{-1}\left(x^2-4\right). \tag{1.2.22.1}$$

Solution: Step 1: Find the domain
For $\sin^{-1}(u)$ to be defined for real values,

$$-1\le u\le 1. \tag{1.2.22.2}$$

Here, $u=x^2-4$. Therefore,

$$-1\le x^2-4\le 1. \tag{1.2.22.3}$$

Adding 4 throughout, we get

$$3\le x^2\le 5. \tag{1.2.22.4}$$

Taking square roots and considering both positive and negative values of x,

$$\sqrt{3} \le |x| \le \sqrt{5}. \tag{1.2.22.5}$$

Hence,

$$-\sqrt{5} \le x \le -\sqrt{3} \tag{1.2.22.6}$$

or

$$\sqrt{3} \le x \le \sqrt{5}. \tag{1.2.22.7}$$

Therefore, the domain is

$$[-\sqrt{5}, -\sqrt{3}] \cup [\sqrt{3}, \sqrt{5}]. \tag{1.2.22.8}$$

Step 2: Find the range
From the domain condition,

$$3 \le x^2 \le 5. \tag{1.2.22.9}$$

Subtracting 4,

$$-1 \le x^2 - 4 \le 1. \tag{1.2.22.10}$$

Thus, the expression $x^2 - 4$ takes every value in the interval $[-1, 1]$.
Since

$$f(x) = \sin^{-1}\left(x^2 - 4\right), \tag{1.2.22.11}$$

and the principal range of $\sin^{-1}(u)$ is

$$-\frac{\pi}{2} \le \sin^{-1}(u) \le \frac{\pi}{2}, \tag{1.2.22.12}$$

the range of $f(x)$ is

$$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]. \tag{1.2.22.13}$$

Final Answer

$$\text{Domain} = [-\sqrt{5}, -\sqrt{3}] \cup [\sqrt{3}, \sqrt{5}] \tag{1.2.22.14}$$

$$\text{Range} = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \tag{1.2.22.15}$$

1.2.23 If $f(x) = |\tan 2x|$,then find the value of $f'(x)$ at $x = \frac{\pi}{3}$.
Solution: Given,

$$f(x) = |\tan 2x|. \tag{1.2.23.1}$$

At $x = \frac{\pi}{3}$,

$$\tan 2x = \tan\left(\frac{2\pi}{3}\right) = -\sqrt{3} < 0. \tag{1.2.23.2}$$

Since $\tan 2x$ is negative near $x = \frac{\pi}{3}$,

$$f(x) = -\tan 2x. \tag{1.2.23.3}$$

Differentiating with respect to x,

$$f'(x) = -2\sec^2 2x. \tag{1.2.23.4}$$

Substituting $x = \frac{\pi}{3}$,

$$f'\left(\frac{\pi}{3}\right) = -2\sec^2\left(\frac{2\pi}{3}\right). \tag{1.2.23.5}$$

We know that

$$\cos\left(\frac{2\pi}{3}\right) = -\frac{1}{2}. \tag{1.2.23.6}$$

Therefore,

$$\sec^2\left(\frac{2\pi}{3}\right) = \frac{1}{\cos^2(2\pi/3)} = \frac{1}{1/4} = 4. \tag{1.2.23.7}$$

Hence,

$$f'\left(\frac{\pi}{3}\right) = -2(4) = -8. \tag{1.2.23.8}$$

$$f'\left(\frac{\pi}{3}\right) = -8 \tag{1.2.23.9}$$

1.2.24 If $y = cosec\left(cot^{-1}x\right)$, then prove that $\left(1+x^2\right)\frac{dy}{dx} - xy = 0$.

Solution: Given, $y = cosec\left(cot^{-1}x\right)$.
Let

$$\theta = cot^{-1}x. \tag{1.2.24.1}$$

Then

$$cot\theta = x. \tag{1.2.24.2}$$

Using the identity

$$cosec^2\theta = 1 + cot^2\theta, \tag{1.2.24.3}$$

we get

$$cosec^2\theta = 1 + x^2. \tag{1.2.24.4}$$

Since $\theta = cot^{-1}x$ lies in $(0, \pi)$, $cosec\theta$ is positive. Therefore,

$$y = \sqrt{1 + x^2}. \tag{1.2.24.5}$$

Squaring both sides,

$$y^2 = 1 + x^2. \tag{1.2.24.6}$$

Differentiating with respect to x,

$$2y\frac{dy}{dx} = 2x. \tag{1.2.24.7}$$

Thus,

$$\frac{dy}{dx} = \frac{x}{y}. \tag{1.2.24.8}$$

Multiplying both sides by $\left(1 + x^2\right)$,

$$\left(1 + x^2\right)\frac{dy}{dx} = \left(1 + x^2\right)\frac{x}{y}. \tag{1.2.24.9}$$

Since $y^2 = 1 + x^2$,

$$\left(1 + x^2\right)\frac{dy}{dx} = \frac{y^2x}{y} = xy. \tag{1.2.24.10}$$

Therefore,

$$\left(1 + x^2\right)\frac{dy}{dx} - xy = xy - xy = 0. \tag{1.2.24.11}$$

Hence proved.

1.2.25 If M and m denote the local maximum and local minimum values of the function $f(x) = x + \frac{1}{x}, \quad x \neq 0$, respectively, find the value of $(M - m)$.

Solution: Given,

$$f(x) = x + \frac{1}{x}. \tag{1.2.25.1}$$

Step 1: Find the first derivative
Differentiating with respect to x,

$$f'(x) = 1 - \frac{1}{x^2}. \tag{1.2.25.2}$$

At the critical points,

$$f'(x) = 0. \tag{1.2.25.3}$$

Therefore,

$$1 - \frac{1}{x^2} = 0. \tag{1.2.25.4}$$

Multiplying by x^2, we get

$$x^2 - 1 = 0. \tag{1.2.25.5}$$

Hence,

$$(x-1)(x+1) = 0. \tag{1.2.25.6}$$

Thus, the critical points are

$$x = 1 \quad \text{or} \quad x = -1. \tag{1.2.25.7}$$

Step 2: Find the second derivative
Differentiating again,

$$f''(x) = \frac{2}{x^3}. \tag{1.2.25.8}$$

At $x = -1$,

$$f''(-1) = \frac{2}{(-1)^3} = -2 < 0. \tag{1.2.25.9}$$

Therefore, $f(x)$ has a local maximum at $x = -1$.
The local maximum value is

$$M = f(-1) = -1 + \frac{1}{-1} = -2. \tag{1.2.25.10}$$

At $x = 1$,

$$f''(1) = \frac{2}{1^3} = 2 > 0. \tag{1.2.25.11}$$

Therefore, $f(x)$ has a local minimum at $x = 1$.
The local minimum value is

$$m = f(1) = 1 + \frac{1}{1} = 2. \tag{1.2.25.12}$$

Step 3: Calculate $(M - m)$
Substituting $M = -2$ and $m = 2$,

$$M - m = -2 - 2. \tag{1.2.25.13}$$

Therefore,

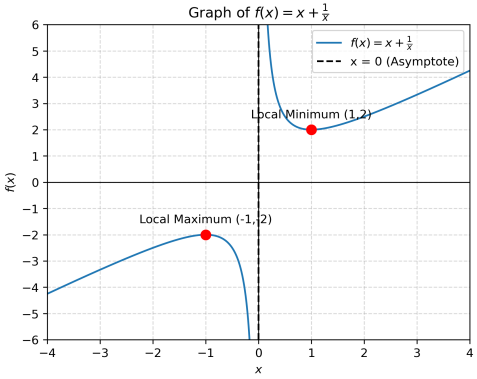
$$M - m = -4. \tag{1.2.25.14}$$

The following code gives the figure

codes/s1225.py

1.2.26 Evaluate:

$$I = \int \frac{e^{4x} - 1}{e^{4x} + 1}\, dx \tag{1.2.26.1}$$



Solution: Rewrite the integrand as

$$I = \int \left(1 - \frac{2}{e^{4x}+1}\right) dx. \tag{1.2.26.2}$$

Therefore,

$$I = x - 2\int \frac{1}{e^{4x}+1}\, dx. \tag{1.2.26.3}$$

Multiply the numerator and denominator of the integrand by e^{-4x}:

$$I = x - 2\int \frac{e^{-4x}}{1+e^{-4x}}\, dx. \tag{1.2.26.4}$$

Let

$$u = 1 + e^{-4x}. \tag{1.2.26.5}$$

Then

$$du = -4e^{-4x}\, dx. \tag{1.2.26.6}$$

Hence by substituting (1.2.26.4) becomes ,

$$I = x - 2\left(-\frac{1}{4}\int \frac{du}{u}\right) \tag{1.2.26.7}$$

(1.2.26.8)

by using (**??**)

$$I = x + \frac{1}{2}\ln|u| + C. \tag{1.2.26.9}$$

Substituting $u = 1 + e^{-4x}$,

$$I = x + \frac{1}{2}\ln\left(1 + e^{-4x}\right) + C. \tag{1.2.26.10}$$

Using

$$1+e^{-4x}=\frac{e^{4x}+1}{e^{4x}}, \tag{1.2.26.11}$$

we obtain

$$I=x+\frac{1}{2}\ln\left(\frac{e^{4x}+1}{e^{4x}}\right)+C \tag{1.2.26.12}$$

$$=x+\frac{1}{2}\ln\left(e^{4x}+1\right)-2x+C \tag{1.2.26.13}$$

$$=\frac{1}{2}\ln\left(e^{4x}+1\right)-x+C. \tag{1.2.26.14}$$

1.2.27 Show that $f(x)=e^{x}-e^{-x}+x-\tan^{-1}x$ is strictly increasing in its domain.
Solution: Given,

$$f(x)=e^{x}-e^{-x}+x-\tan^{-1}x. \tag{1.2.27.1}$$

The domain of the function is $\mathbb{R}$.
Step 1: Differentiate the function
Differentiating with respect to x, we get

$$f'(x)=e^{x}+e^{-x}+1-\frac{1}{1+x^{2}}. \tag{1.2.27.2}$$

Rearranging,

$$f'(x)=e^{x}+e^{-x}+\left(1-\frac{1}{1+x^{2}}\right). \tag{1.2.27.3}$$

Simplifying the expression in parentheses,

$$f'(x)=e^{x}+e^{-x}+\frac{x^{2}}{1+x^{2}}. \tag{1.2.27.4}$$

Step 2: Prove that $f'(x)>0$
We know that, for every real number x,

$$e^{x}+e^{-x}=2. \tag{1.2.27.5}$$

Also, since $x^{2}=0$,

$$\frac{x^{2}}{1+x^{2}}=0. \tag{1.2.27.6}$$

Therefore,

$$f'(x)=e^{x}+e^{-x}+\frac{x^{2}}{1+x^{2}}=2. \tag{1.2.27.7}$$

Hence,

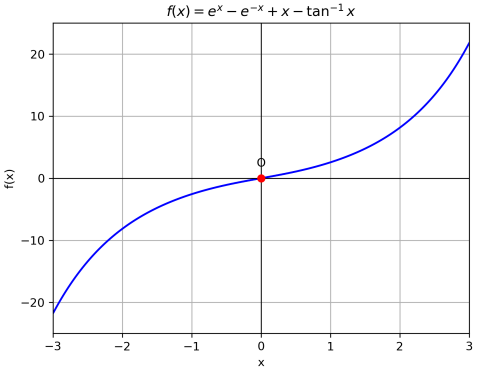
$$f'(x)=2>0 \qquad \forall x\in\mathbb{R}. \tag{1.2.27.8}$$

Step 3: Conclusion
Since

$$f'(x) > 0 \qquad \forall x \in \mathbb{R}, \tag{1.2.27.9}$$

the function $f(x)$ is strictly increasing throughout its domain. The following code gives the figure

codes/s1227.py



1.2.28 If $x = e^{\cos 3t}$ and $y = e^{\sin 3t}$, prove that $\frac{dy}{dx} = -\frac{y \log x}{x \log y}$.
Solution: Given,

$$x = e^{\cos 3t} \qquad \text{and} \qquad y = e^{\sin 3t}. \tag{1.2.28.1}$$

Taking logarithms on both sides, we get

$$\log x = \cos 3t \tag{1.2.28.2}$$

and

$$\log y = \sin 3t. \tag{1.2.28.3}$$

Step 1: Differentiate x with respect to t
Given,

$$x = e^{\cos 3t}. \tag{1.2.28.4}$$

Differentiating with respect to t using the formulae (1.1.13),

$$\frac{dx}{dt} = e^{\cos 3t}\left(-3 \sin 3t\right). \tag{1.2.28.5}$$

Since $x = e^{\cos 3t}$ and $\sin 3t = \log y$,

$$\frac{dx}{dt} = -3x \log y. \tag{1.2.28.6}$$

Step 2: Differentiate y with respect to t using the
Given,

$$y = e^{\sin 3t}. \tag{1.2.28.7}$$

Differentiating with respect to t using the formulae (1.1.13),,

$$\frac{dy}{dt} = e^{\sin 3t} \left(3 \cos 3t\right). \tag{1.2.28.8}$$

Since $y = e^{\sin 3t}$ and $\cos 3t = \log x$,

$$\frac{dy}{dt} = 3y \log x. \tag{1.2.28.9}$$

Step 3: Calculate $\frac{dy}{dx}$
Using the formula for parametric differentiation,

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}. \tag{1.2.28.10}$$

Substituting the values obtained above,

$$\frac{dy}{dx} = \frac{3y \log x}{-3x \log y}. \tag{1.2.28.11}$$

Cancelling 3, we get

$$\frac{dy}{dx} = -\frac{y \log x}{x \log y}. \tag{1.2.28.12}$$

Hence proved.

1.2.29 Show that

$$\frac{d}{dx}\left(|x|\right) = \frac{|x|}{x}, \qquad x \neq 0. \tag{1.2.29.1}$$

Solution: We know that the modulus function is defined as

$$|x| = \begin{cases} x, & x > 0, \\ -x, & x < 0. \end{cases} \tag{1.2.29.2}$$

Case 1: When $x > 0$
Since $|x| = x$,

$$\frac{d}{dx}\left(|x|\right) = \frac{d}{dx}\left(x\right) = 1. \tag{1.2.29.3}$$

Also, for $x > 0$,

$$\frac{|x|}{x} = \frac{x}{x} = 1. \tag{1.2.29.4}$$

Therefore,

$$\frac{d}{dx}\left(|x|\right) = \frac{|x|}{x}. \tag{1.2.29.5}$$

Case 2: When $x < 0$
Since $|x| = -x$,

$$\frac{d}{dx}\left(|x|\right) = \frac{d}{dx}\left(-x\right) = -1. \tag{1.2.29.6}$$

Also, for $x < 0$,

$$\frac{|x|}{x} = \frac{-x}{x} = -1. \tag{1.2.29.7}$$

Therefore,

$$\frac{d}{dx}\left(|x|\right) = \frac{|x|}{x}. \tag{1.2.29.8}$$

Thus, in both cases,

$$\frac{d}{dx}\left(|x|\right) = \frac{|x|}{x}, \qquad x \neq 0. \tag{1.2.29.9}$$

Hence proved.

1.2.30 Evaluate:

$$I = \int_{-2}^{2} \sqrt{\frac{2+x}{2-x}}\, dx \tag{1.2.30.1}$$

Solution: Let

$$x = 2\cos\theta. \tag{1.2.30.2}$$

Then,

$$dx = -2\sin\theta\, d\theta. \tag{1.2.30.3}$$

Also,

$$\sqrt{\frac{2+x}{2-x}} = \sqrt{\frac{2+2\cos\theta}{2-2\cos\theta}}. \tag{1.2.30.4}$$

Using the identities

$$1 + \cos\theta = 2\cos^2\frac{\theta}{2} \tag{1.2.30.5}$$

and

$$1 - \cos\theta = 2\sin^2\frac{\theta}{2}, \tag{1.2.30.6}$$

we obtain

$$\sqrt{\frac{2+x}{2-x}} = \cot\frac{\theta}{2}. \tag{1.2.30.7}$$

Changing the limits:
When $x = -2$,

$$-2 = 2\cos\theta \implies \theta = \pi. \tag{1.2.30.8}$$

When $x = 2$,

$$2 = 2\cos\theta \implies \theta = 0. \tag{1.2.30.9}$$

Therefore,

$$I = \int_{\pi}^{0} \cot\frac{\theta}{2}\left(-2\sin\theta\right)\,d\theta. \tag{1.2.30.10}$$

Reversing the limits,

$$I = 2\int_{0}^{\pi} \cot\frac{\theta}{2}\sin\theta\,d\theta. \tag{1.2.30.11}$$

Using

$$\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}, \tag{1.2.30.12}$$

we get

$$I = 4\int_{0}^{\pi} \cos^2\frac{\theta}{2}\,d\theta. \tag{1.2.30.13}$$

Now, using

$$\cos^2\frac{\theta}{2} = \frac{1+\cos\theta}{2}, \tag{1.2.30.14}$$

we obtain

$$I = 4\int_{0}^{\pi} \frac{1+\cos\theta}{2}\,d\theta. \tag{1.2.30.15}$$

Hence,

$$I = 2\int_{0}^{\pi} \left(1+\cos\theta\right)\,d\theta. \tag{1.2.30.16}$$

Integrating,

$$I = 2\left[\theta + \sin\theta\right]_{0}^{\pi}. \tag{1.2.30.17}$$

Substituting the limits,

$$I = 2\left[(\pi+0) - (0+0)\right]. \tag{1.2.30.18}$$

Thus,

$$I = 2\pi \tag{1.2.30.19}$$

1.2.31 Evaluate:

$$I = \int \frac{1}{x\left[(\log x)^2 - 3\log x - 4\right]}\, dx. \tag{1.2.31.1}$$

Solution:
Let

$$t = \log x. \tag{1.2.31.2}$$

Then,

$$dt = \frac{1}{x}\, dx. \tag{1.2.31.3}$$

Therefore,

$$I = \int \frac{1}{t^2 - 3t - 4}\, dt. \tag{1.2.31.4}$$

Factorising the denominator,

$$t^2 - 3t - 4 = (t-4)(t+1). \tag{1.2.31.5}$$

Hence,

$$I = \int \frac{1}{(t-4)(t+1)}\, dt. \tag{1.2.31.6}$$

Using partial fractions, let

$$\frac{1}{(t-4)(t+1)} = \frac{A}{t-4} + \frac{B}{t+1}. \tag{1.2.31.7}$$

Multiplying both sides by $(t-4)(t+1)$,

$$1 = A(t+1) + B(t-4). \tag{1.2.31.8}$$

Putting $t = 4$,

$$1 = 5A \implies A = \frac{1}{5}. \tag{1.2.31.9}$$

Putting $t = -1$,

$$1 = -5B \implies B = -\frac{1}{5}. \tag{1.2.31.10}$$

Therefore substituting A nad B in (1.2.31.7) we get,

$$I = \frac{1}{5}\int \frac{1}{t-4}\, dt - \frac{1}{5}\int \frac{1}{t+1}\, dt. \tag{1.2.31.11}$$

Integrating,

$$I = \frac{1}{5}\log|t-4| - \frac{1}{5}\log|t+1| + C. \tag{1.2.31.12}$$

Combining the logarithms,

$$I = \frac{1}{5}\log\left|\frac{t-4}{t+1}\right| + C. \tag{1.2.31.13}$$

Substituting $t = \log x$, we obtain

$$I = \frac{1}{5}\log\left|\frac{\log x - 4}{\log x + 1}\right| + C \tag{1.2.31.14}$$

1.2.32 Find the particular solution of the differential equation

$$2xy + y^2 - 2x^2\frac{dy}{dx} = 0, \tag{1.2.32.1}$$

given that $y = 2$ when $x = 1$.

Solution:

Rearranging (1.2.32.1),

$$\frac{dy}{dx} = \frac{y}{x} + \frac{y^2}{2x^2}. \tag{1.2.32.2}$$

Put

$$z = \frac{1}{y}. \tag{1.2.32.3}$$

Differentiating,

$$\frac{dz}{dx} = -\frac{1}{y^2}\frac{dy}{dx}. \tag{1.2.32.4}$$

Substituting from (1.2.32.2),

$$\frac{dz}{dx} = -\frac{z}{x} - \frac{1}{2x^2}. \tag{1.2.32.5}$$

Therefore,

$$\frac{dz}{dx} + \frac{z}{x} = -\frac{1}{2x^2}. \tag{1.2.32.6}$$

Multiplying throughout by x,

$$x\frac{dz}{dx} + z = -\frac{1}{2x} \tag{1.2.32.7}$$

$$\frac{d}{dx}(xz) = -\frac{1}{2x}. \tag{1.2.32.8}$$

Let

$$t = x - 1, \qquad w(t) = xz = \frac{x}{y}. \tag{1.2.32.9}$$

Then $x = t + 1$, and

$$\frac{dw}{dt} = -\frac{1}{2(t+1)}. \tag{1.2.32.10}$$

The initial condition gives

$$w(0) = \frac{1}{2}. \tag{1.2.32.11}$$

Taking the Laplace transform of (1.2.32.10), and writing $W(s) = \mathcal{L}\{w(t)\}$, use the derivative property (1.1.17.2) and the transform in (1.1.17.7):

$$sW(s) - w(0) = -\frac{1}{2}e^s E_1(s). \tag{1.2.32.12}$$

Using (1.2.32.11),

$$sW(s) - \frac{1}{2} = -\frac{1}{2}e^s E_1(s). \tag{1.2.32.13}$$

Hence,

$$W(s) = \frac{1}{2s} - \frac{e^s E_1(s)}{2s}. \tag{1.2.32.14}$$

Using the transform pair in (1.1.17.8),

$$w(t) = \frac{1}{2} - \frac{1}{2}\ln(1+t). \tag{1.2.32.15}$$

From (1.2.32.9), $t = x - 1$ and $w = x/y$. Therefore,

$$\frac{x}{y} = \frac{1}{2} - \frac{1}{2}\ln x \tag{1.2.32.16}$$

$$\frac{2x}{y} = 1 - \ln x. \tag{1.2.32.17}$$

Thus, the particular solution is

$$y = \frac{2x}{1 - \ln x}. \tag{1.2.32.18}$$

1.2.33 Find the general solution of the differential equation

$$y\,dx = (x + 2y^2)\,dy. \tag{1.2.33.1}$$

Solution:
Dividing by dy and rearranging,

$$y\frac{dx}{dy} - x = 2y^2. \tag{1.2.33.2}$$

Let

$$X(s) = \mathcal{L}\{x(y)\}, \tag{1.2.33.3}$$

where the Laplace transform is taken with respect to y.

Using the multiplication property (1.1.17.3) and the derivative property (1.1.17.2),

$$\mathcal{L}\left\{y\frac{dx}{dy}\right\} = -\frac{d}{ds}\mathcal{L}\left\{\frac{dx}{dy}\right\} \tag{1.2.33.4}$$

$$= -\frac{d}{ds}\left(sX(s) - x(0)\right) \tag{1.2.33.5}$$

$$= -X(s) - s\frac{dX}{ds}. \tag{1.2.33.6}$$

Taking the Laplace transform of (1.2.33.2), and using (1.1.17.5) with $n = 2$,

$$-X(s) - s\frac{dX}{ds} - X(s) = \frac{4}{s^3}. \tag{1.2.33.7}$$

Therefore,

$$s\frac{dX}{ds} + 2X(s) = -\frac{4}{s^3}. \tag{1.2.33.8}$$

Dividing by s,

$$\frac{dX}{ds} + \frac{2}{s}X(s) = -\frac{4}{s^4}. \tag{1.2.33.9}$$

The integrating factor is

$$\text{I.F.} = e^{\int \frac{2}{s}\,ds} = s^2. \tag{1.2.33.10}$$

Multiplying throughout by s^2,

$$s^2\frac{dX}{ds} + 2sX(s) = -\frac{4}{s^2} \tag{1.2.33.11}$$

$$\frac{d}{ds}\left(s^2X(s)\right) = -\frac{4}{s^2}. \tag{1.2.33.12}$$

Integrating with respect to s,

$$s^2X(s) = \frac{4}{s} + C. \tag{1.2.33.13}$$

Hence,

$$X(s) = \frac{4}{s^3} + \frac{C}{s^2}. \tag{1.2.33.14}$$

Taking the inverse Laplace transform using (1.1.17.9),

$$x(y) = 4\frac{y^2}{2!} + Cy \tag{1.2.33.15}$$

$$= 2y^2 + Cy. \tag{1.2.33.16}$$

Therefore, the general solution is

$$x = 2y^2 + Cy. \tag{1.2.33.17}$$

1.2.34 The position vectors of the vertices A, B and C of a triangle ABC are $\mathbf{A} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}, \mathbf{B} = \mathbf{i} - 3\mathbf{j} - 5\mathbf{k}$, and $\mathbf{C} = 3\mathbf{i} - 4\mathbf{j} - 4\mathbf{k}$. Find all the angles of triangle

ABC.

Solution: The position vectors are

$$A = (2, -1, 1), \quad B = (1, -3, -5), \quad C = (3, -4, -4). \tag{1.2.34.1}$$

Step 1: Find the vectors along the sides.
We have

$$\overrightarrow{AB} = \mathbf{B} - \mathbf{A}. \tag{1.2.34.2}$$

Therefore,

$$\overrightarrow{AB} = (1-2)\,\mathbf{i} + (-3+1)\,\mathbf{j} + (-5-1)\,\mathbf{k}. \tag{1.2.34.3}$$

Hence,

$$\overrightarrow{AB} = -\mathbf{i} - 2\mathbf{j} - 6\mathbf{k}. \tag{1.2.34.4}$$

Similarly,

$$\overrightarrow{AC} = \mathbf{C} - \mathbf{A}, \tag{1.2.34.5}$$

which gives

$$\overrightarrow{AC} = \mathbf{i} - 3\mathbf{j} - 5\mathbf{k}. \tag{1.2.34.6}$$

Also,

$$\overrightarrow{BC} = \mathbf{C} - \mathbf{B}, \tag{1.2.34.7}$$

so

$$\overrightarrow{BC} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}. \tag{1.2.34.8}$$

Step 2: Find angle A.
The angle A is the angle between $\overrightarrow{AB}$ and $\overrightarrow{AC}$.
Using the dot product formula,

$$\cos A = \frac{\overrightarrow{AB} \cdot \overrightarrow{AC}}{|\overrightarrow{AB}||\overrightarrow{AC}|}. \tag{1.2.34.9}$$

First, calculate the dot product:

$$\overrightarrow{AB} \cdot \overrightarrow{AC} = (-1)(1) + (-2)(-3) + (-6)(-5). \tag{1.2.34.10}$$

Thus,

$$\overrightarrow{AB} \cdot \overrightarrow{AC} = -1 + 6 + 30 = 35. \tag{1.2.34.11}$$

The magnitudes are

$$|\overrightarrow{AB}| = \sqrt{(-1)^2 + (-2)^2 + (-6)^2} = \sqrt{41}, \tag{1.2.34.12}$$

and

$$|\overrightarrow{AC}| = \sqrt{1^2 + (-3)^2 + (-5)^2} = \sqrt{35}. \tag{1.2.34.13}$$

Therefore,

$$\cos A = \frac{35}{\sqrt{41}\sqrt{35}} = \sqrt{\frac{35}{41}}. \tag{1.2.34.14}$$

Hence,

$$A = \cos^{-1}\sqrt{\frac{35}{41}}. \tag{1.2.34.15}$$

Step 3: Find angle B.
The angle B is the angle between $\overrightarrow{BA}$ and $\overrightarrow{BC}$. We have

$$\overrightarrow{BA} = \mathbf{i} + 2\mathbf{j} + 6\mathbf{k}. \tag{1.2.34.16}$$

Using the dot product formula (1.1.7.1),

$$\cos B = \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{|\overrightarrow{BA}||\overrightarrow{BC}|}. \tag{1.2.34.17}$$

Calculating the dot product,

$$\overrightarrow{BA} \cdot \overrightarrow{BC} = (1)(2) + (2)(-1) + (6)(1). \tag{1.2.34.18}$$

Thus,

$$\overrightarrow{BA} \cdot \overrightarrow{BC} = 2 - 2 + 6 = 6. \tag{1.2.34.19}$$

Also,

$$|\overrightarrow{BA}| = \sqrt{41}, \tag{1.2.34.20}$$

and

$$|\overrightarrow{BC}| = \sqrt{2^2 + (-1)^2 + 1^2} = \sqrt{6}. \tag{1.2.34.21}$$

Therefore,

$$\cos B = \frac{6}{\sqrt{41}\sqrt{6}} = \sqrt{\frac{6}{41}}. \tag{1.2.34.22}$$

Hence,

$$B = \cos^{-1}\sqrt{\frac{6}{41}} \tag{1.2.34.23}$$

Step 4: Find angle C.
The angle C is the angle between $\overrightarrow{CA}$ and $\overrightarrow{CB}$.

We have

$$\overrightarrow{CA} = -\mathbf{i} + 3\mathbf{j} + 5\mathbf{k}, \tag{1.2.34.24}$$

and

$$\overrightarrow{CB} = -2\mathbf{i} + \mathbf{j} - \mathbf{k}. \tag{1.2.34.25}$$

Their dot product is

$$\overrightarrow{CA} \cdot \overrightarrow{CB} = (-1)(-2) + (3)(1) + (5)(-1). \tag{1.2.34.26}$$

Therefore,

$$\overrightarrow{CA} \cdot \overrightarrow{CB} = 2 + 3 - 5 = 0. \tag{1.2.34.27}$$

Since the dot product is zero, the vectors are perpendicular.
Hence,

$$C = 90^\circ \tag{1.2.34.28}$$

Final Answer

The three angles of triangle ABC are

$$A = \cos^{-1}\sqrt{\frac{35}{41}}, \tag{1.2.34.29}$$

$$B = \cos^{-1}\sqrt{\frac{6}{41}}, \tag{1.2.34.30}$$

$$C = 90^\circ. \tag{1.2.34.31}$$

1.2.35 A pair of dice is thrown simultaneously. If X denotes the absolute difference of the numbers appearing on top of the dice,then find the probability distribution of X.
Solution: When two dice are thrown simultaneously, the total number of possible outcomes is

$$n(S) = 6 \times 6 = 36. \tag{1.2.35.1}$$

Let the numbers appearing on the two dice be a and b.
The random variable X is defined as

$$X = |a - b|. \tag{1.2.35.2}$$

Since the numbers on the dice range from 1 to 6, the possible values of X are

$$X = 0, 1, 2, 3, 4, 5. \tag{1.2.35.3}$$

Step 1: Calculate the number of outcomes for each value of X.
Case 1: $X = 0$
Both dice show the same number.

The outcomes are

$$(1,1),(2,2),(3,3),(4,4),(5,5),(6,6). \tag{1.2.35.4}$$

Therefore,

$$P(X=0)=\frac{6}{36}=\frac{1}{6}. \tag{1.2.35.5}$$

Case 2: $X=1$
The outcomes are

$$(1,2),(2,1),(2,3),(3,2), \tag{1.2.35.6}$$
$$(3,4),(4,3),(4,5),(5,4), \tag{1.2.35.7}$$
$$(5,6),(6,5). \tag{1.2.35.8}$$

There are 10 outcomes. Hence,

$$P(X=1)=\frac{10}{36}=\frac{5}{18}. \tag{1.2.35.9}$$

Case 3: $X=2$
There are 8 outcomes:

$$(1,3),(3,1),(2,4),(4,2), \tag{1.2.35.10}$$
$$(3,5),(5,3),(4,6),(6,4). \tag{1.2.35.11}$$

Therefore,

$$P(X=2)=\frac{8}{36}=\frac{2}{9}. \tag{1.2.35.12}$$

Case 4: $X=3$
There are 6 outcomes:

$$(1,4),(4,1),(2,5),(5,2),(3,6),(6,3). \tag{1.2.35.13}$$

Therefore,

$$P(X=3)=\frac{6}{36}=\frac{1}{6}. \tag{1.2.35.14}$$

Case 5: $X=4$ There are 4 outcomes:

$$(1,5),(5,1),(2,6),(6,2). \tag{1.2.35.15}$$

Therefore,

$$P(X=4)=\frac{4}{36}=\frac{1}{9}. \tag{1.2.35.16}$$

Case 6: $X=5$
There are 2 outcomes:

$$(1,6),(6,1). \tag{1.2.35.17}$$

Therefore,

$$P(X=5)=\frac{2}{36}=\frac{1}{18}. \tag{1.2.35.18}$$

Step 2: Construct the probability distribution.
The probability distribution of X is given by

X	0	1	2	3	4	5
$P(X)$	$\frac{1}{6}$	$\frac{5}{18}$	$\frac{2}{9}$	$\frac{1}{6}$	$\frac{1}{9}$	$\frac{1}{18}$

(1.2.35.19)

Verification:
The sum of all probabilities must equal 1.

$$\sum P(X)=\frac{1}{6}+\frac{5}{18}+\frac{2}{9}+\frac{1}{6}+\frac{1}{9}+\frac{1}{18} \tag{1.2.35.20}$$

$$=\frac{3+5+4+3+2+1}{18} \tag{1.2.35.21}$$

$$=\frac{18}{18} \tag{1.2.35.22}$$

$$=1. \tag{1.2.35.23}$$

Hence, the probability distribution is correct.

1.2.36 Evaluate:$I=\int x^2\sin^{-1}\left(x^{3/2}\right)\,dx$.

Solution: Using integration by parts (1.1.3.1),

$$\int u\,dv=uv-\int v\,du. \tag{1.2.36.1}$$

Choose

$$u=\sin^{-1}\left(x^{3/2}\right),\qquad dv=x^2\,dx. \tag{1.2.36.2}$$

Differentiating u using the formulae (1.1.15),

$$du=\frac{1}{\sqrt{1-\left(x^{3/2}\right)^2}}\cdot\frac{3}{2}x^{1/2}\,dx. \tag{1.2.36.3}$$

Therefore,

$$du=\frac{3\sqrt{x}}{2\sqrt{1-x^3}}\,dx. \tag{1.2.36.4}$$

Integrating dv,

$$v=\int x^2\,dx=\frac{x^3}{3}. \tag{1.2.36.5}$$

Applying integration by parts,

$$I=\frac{x^3}{3}\sin^{-1}\left(x^{3/2}\right)-\int\frac{x^3}{3}\cdot\frac{3\sqrt{x}}{2\sqrt{1-x^3}}\,dx. \tag{1.2.36.6}$$

Simplifying,

$$I = \frac{x^3}{3}\sin^{-1}\left(x^{3/2}\right) - \frac{1}{2}\int \frac{x^{7/2}}{\sqrt{1-x^3}}\,dx. \tag{1.2.36.7}$$

Let

$$t = x^{3/2}. \tag{1.2.36.8}$$

Then

$$dt = \frac{3}{2}\sqrt{x}\,dx, \qquad dx = \frac{2}{3\sqrt{x}}\,dt. \tag{1.2.36.9}$$

Since $x = t^{2/3}$, the remaining integral becomes

$$\int \frac{x^{7/2}}{\sqrt{1-x^3}}\,dx = \frac{2}{3}\int \frac{t^2}{\sqrt{1-t^2}}\,dt. \tag{1.2.36.10}$$

Using

$$t = \sin\theta, \qquad dt = \cos\theta\,d\theta, \tag{1.2.36.11}$$

we obtain

$$\int \frac{t^2}{\sqrt{1-t^2}}\,dt = \int \sin^2\theta\,d\theta. \tag{1.2.36.12}$$

Using the identity

$$\sin^2\theta = \frac{1-\cos 2\theta}{2}, \tag{1.2.36.13}$$

we get

$$\int \sin^2\theta\,d\theta = \frac{\theta}{2} - \frac{\sin 2\theta}{4} + C. \tag{1.2.36.14}$$

Substituting back,

$$\int \frac{t^2}{\sqrt{1-t^2}}\,dt = \frac{1}{2}\sin^{-1} t - \frac{t\sqrt{1-t^2}}{2} + C. \tag{1.2.36.15}$$

Therefore,

$$\int \frac{x^{7/2}}{\sqrt{1-x^3}}\,dx = \frac{1}{3}\sin^{-1}\left(x^{3/2}\right) - \frac{x^{3/2}\sqrt{1-x^3}}{3} + C. \tag{1.2.36.16}$$

Substituting into the expression for I,

$$I = \frac{x^3}{3}\sin^{-1}\left(x^{3/2}\right) \tag{1.2.36.17}$$

$$- \frac{1}{6}\sin^{-1}\left(x^{3/2}\right) \tag{1.2.36.18}$$

$$+ \frac{x^{3/2}\sqrt{1-x^3}}{6} + C. \tag{1.2.36.19}$$

Hence, the final answer is

$$I = \frac{x^3}{3}\sin^{-1}\left(x^{3/2}\right) - \frac{1}{6}\sin^{-1}\left(x^{3/2}\right) \tag{1.2.36.20}$$

$$+ \frac{x^{3/2}\sqrt{1-x^3}}{6} + C. \tag{1.2.36.21}$$

1.2.37 Show that the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \dfrac{2x}{1+x^2}$ is neither one-one nor onto. Further, find a set A so that the given function $f : \mathbb{R} \to A$ becomes onto function.

Solution: The given function is

$$f(x) = \frac{2x}{1+x^2}. \tag{1.2.37.1}$$

Step 1: Show that f is not one-one.
A function is one-one if distinct inputs always have distinct outputs.
Consider

$$f(2) = \frac{2(2)}{1+2^2} = \frac{4}{5}. \tag{1.2.37.2}$$

Also,

$$f\left(\frac{1}{2}\right) = \frac{2\left(\frac{1}{2}\right)}{1+\left(\frac{1}{2}\right)^2}. \tag{1.2.37.3}$$

Simplifying,

$$f\left(\frac{1}{2}\right) = \frac{1}{1+\frac{1}{4}} = \frac{4}{5}. \tag{1.2.37.4}$$

Therefore,

$$f(2) = f\left(\frac{1}{2}\right), \tag{1.2.37.5}$$

even though

$$2 \neq \frac{1}{2}. \tag{1.2.37.6}$$

Hence, f is not one-one.

$$f \text{ is not one-one.} \tag{1.2.37.7}$$

Step 2: Show that f is not onto $\mathbb{R}$.
We have

$$f(x) = \frac{2x}{1+x^2}. \tag{1.2.37.8}$$

Since

$$(|x|-1)^2 = 0, \tag{1.2.37.9}$$

we get

$$x^2 + 1 = 2|x|. \tag{1.2.37.10}$$

Therefore,

$$\left| \frac{2x}{1+x^2} \right| \leq 1. \tag{1.2.37.11}$$

Thus,

$$-1 \leq f(x) \leq 1. \tag{1.2.37.12}$$

Hence, the range of f is contained in $[-1, 1]$. In fact, both endpoints are attained:

$$f(1) = 1, \qquad f(-1) = -1. \tag{1.2.37.13}$$

Therefore, the range is

$$\text{Range}(f) = [-1, 1]. \tag{1.2.37.14}$$

Since the codomain is $\mathbb{R}$, values outside $[-1, 1]$ are not attained. Hence, f is not onto $\mathbb{R}$.

$$f \text{ is not onto } \mathbb{R}. \tag{1.2.37.15}$$

Step 3: Find the set A.
For $f : \mathbb{R} \to A$ to be onto, its codomain must equal its range. Therefore,

$$A = [-1, 1]. \tag{1.2.37.16}$$

Thus, $f : \mathbb{R} \to [-1, 1]$ is onto.

1.2.38 A relation $\mathbb{R}$ is defined on $\mathbb{N} \times \mathbb{N}$ (where$\mathbb{N}$is the set of natural numbers) as:

$$(a, b)\ R\ (c, d) \iff a - c = b - d. \tag{1.2.38.1}$$

Show that R is an equivalence relation.
Solution: A relation is an equivalence relation if it is reflexive, symmetric, and transitive.
We prove these three properties.
Step 1: Reflexivity
For any $(a, b) \in \mathbb{N} \times \mathbb{N}$, we have

$$a - a = 0 \tag{1.2.38.2}$$

and

$$b - b = 0. \tag{1.2.38.3}$$

Therefore,

$$a - a = b - b. \tag{1.2.38.4}$$

By the definition of R,

$$(a,b)\ R\ (a,b)\,. \tag{1.2.38.5}$$

Hence, R is reflexive.

$$R \text{ is reflexive.} \tag{1.2.38.6}$$

Step 2: Symmetry
Suppose

$$(a,b)\ R\ (c,d)\,. \tag{1.2.38.7}$$

Then, by definition,

$$a-c=b-d. \tag{1.2.38.8}$$

Multiplying both sides by -1, we obtain

$$c-a=d-b. \tag{1.2.38.9}$$

Therefore,

$$(c,d)\ R\ (a,b)\,. \tag{1.2.38.10}$$

Hence, R is symmetric.

$$R \text{ is symmetric.} \tag{1.2.38.11}$$

Step 3: Transitivity
Suppose

$$(a,b)\ R\ (c,d) \tag{1.2.38.12}$$

and

$$(c,d)\ R\ (e,f)\,. \tag{1.2.38.13}$$

Then,

$$a-c=b-d \tag{1.2.38.14}$$

and

$$c-e=d-f. \tag{1.2.38.15}$$

Adding these equations, we get

$$(a-c)+(c-e)=(b-d)+(d-f)\,. \tag{1.2.38.16}$$

Simplifying,

$$a-e=b-f. \tag{1.2.38.17}$$

Therefore,

$$(a, b)\ R\ (e, f)\,. \tag{1.2.38.18}$$

Hence, R is transitive.

$$R \text{ is transitive.} \tag{1.2.38.19}$$

Since R is reflexive, symmetric, and transitive, it follows that

$$R \text{ is an equivalence relation.} \tag{1.2.38.20}$$

1.2.39 Find the equation of the line which bisects the line segment joining $A\,(2, 3, 4)$ and $B\,(4, 5, 8)$ and is perpendicular to the lines $\dfrac{x-8}{3} = \dfrac{y+19}{-16} = \dfrac{z-10}{7}$ and $\dfrac{x-15}{3} = \dfrac{y-29}{8} = \dfrac{z+5}{-5}$.

Solution: The required line passes through the midpoint of AB and is perpendicular to both given lines.

Step 1: Find the midpoint of AB.

The coordinates of the midpoint M of AB are

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right). \tag{1.2.39.1}$$

Substituting $A\,(2, 3, 4)$ and $B\,(4, 5, 8)$,

$$M = \left(\frac{2+4}{2}, \frac{3+5}{2}, \frac{4+8}{2}\right). \tag{1.2.39.2}$$

Therefore,

$$M = (3, 4, 6)\,. \tag{1.2.39.3}$$

Step 2: Find the direction vectors of the given lines.

The first line is

$$\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7}. \tag{1.2.39.4}$$

Its direction vector is

$$\mathbf{d}_1 = 3\mathbf{i} - 16\mathbf{j} + 7\mathbf{k}. \tag{1.2.39.5}$$

Thus,

$$\mathbf{d}_1 = (3, -16, 7)\,. \tag{1.2.39.6}$$

The second line is

$$\frac{x-15}{3} = \frac{y-29}{8} = \frac{z+5}{-5}. \tag{1.2.39.7}$$

Its direction vector is

$$\mathbf{d}_2 = 3\mathbf{i} + 8\mathbf{j} - 5\mathbf{k}. \tag{1.2.39.8}$$

Thus,

$$\mathbf{d}_2 = (3, 8, -5)\,. \tag{1.2.39.9}$$

Step 3: Find the direction vector of the required line.
Since the required line is perpendicular to both given lines, its direction vector must be perpendicular to both $\mathbf{d}_1$ and $\mathbf{d}_2$.
Therefore, we take their cross product:

$$\mathbf{d} = \mathbf{d}_1 \times \mathbf{d}_2. \tag{1.2.39.10}$$

solving the cross product using(1.1.11.1)

$$\mathbf{d} = \begin{pmatrix} 16\,(5) - 7\,(8) \\ 7\,(3) - 3\,(-5) \\ 3\,(8) - (-16)\,3 \end{pmatrix} = \begin{pmatrix} 24 \\ 36 \\ 72 \end{pmatrix} \tag{1.2.39.11}$$

Simplifying,

$$\mathbf{d} = 24\mathbf{i} + 36\mathbf{j} + 72\mathbf{k} \tag{1.2.39.12}$$

$$= 12\,(2\mathbf{i} + 3\mathbf{j} + 6\mathbf{k})\,. \tag{1.2.39.13}$$

We may therefore take the direction vector as

$$\mathbf{d} = (2, 3, 6) \tag{1.2.39.14}$$

Step 4: Write the equation of the required line.
The line passes through $M\,(3, 4, 6)$ and has direction vector $(2, 3, 6)$.
The symmetric equation of a line passing through (x_1, y_1, z_1) with direction ratios (a, b, c) is

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}. \tag{1.2.39.15}$$

Substituting the midpoint and direction ratios, we obtain

$$\frac{x - 3}{2} = \frac{y - 4}{3} = \frac{z - 6}{6} \tag{1.2.39.16}$$

The following code gives the figure

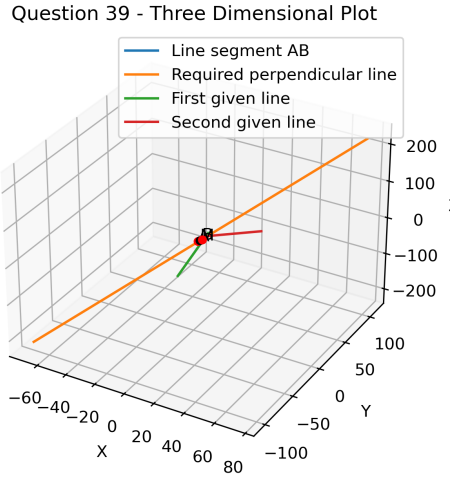
codes/s1239.py

1.2.40 Solve the following system of equations using matrices.

$$\frac{2}{x} + \frac{3}{y} + \frac{10}{z} = 4 \tag{1.2.40.1}$$

$$\frac{4}{x} - \frac{6}{y} + \frac{5}{z} = 1 \tag{1.2.40.2}$$

$$\frac{6}{x} + \frac{9}{y} - \frac{20}{z} = 2 \tag{1.2.40.3}$$



Solution: Let

$$a = \frac{1}{x}, \qquad b = \frac{1}{y}, \qquad c = \frac{1}{z} \tag{1.2.40.4}$$

The equations become

$$2a + 3b + 10c = 4 \tag{1.2.40.5}$$
$$4a - 6b + 5c = 1 \tag{1.2.40.6}$$
$$6a + 9b - 20c = 2 \tag{1.2.40.7}$$

In matrix form,

$$AX = B \tag{1.2.40.8}$$

where

$$A = \begin{pmatrix} 2 & 3 & 10 \\ 4 & -6 & 5 \\ 6 & 9 & -20 \end{pmatrix} \tag{1.2.40.9}$$

$$X = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \tag{1.2.40.10}$$

$$B = \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix} \tag{1.2.40.11}$$

The augmented matrix is

$$\left[\begin{array}{ccc|c} 2 & 3 & 10 & 4 \\ 4 & -6 & 5 & 1 \\ 6 & 9 & -20 & 2 \end{array}\right] \tag{1.2.40.12}$$

Applying the row operations,

$$R_2 \to R_2 - 2R_1 \tag{1.2.40.13}$$
$$R_3 \to R_3 - 3R_1 \tag{1.2.40.14}$$

We get,

$$\left[\begin{array}{ccc|c} 2 & 3 & 10 & 4 \\ 0 & -12 & -15 & -7 \\ 0 & 0 & -50 & -10 \end{array}\right] \tag{1.2.40.15}$$

From the third row,

$$-50c = -10 \tag{1.2.40.16}$$
$$c = \frac{1}{5} \tag{1.2.40.17}$$

From the second row,

$$-12b - 15c = -7 \tag{1.2.40.18}$$

substituting (1.2.40.17)

$$-12b - 15\left(\frac{1}{5}\right) = -7 \tag{1.2.40.19}$$
$$-12b - 3 = -7 \tag{1.2.40.20}$$
$$b = \frac{1}{3} \tag{1.2.40.21}$$

From the first row,

$$2a + 3b + 10c = 4 \tag{1.2.40.22}$$

substituting (1.2.40.17) and (1.2.40.21)

$$2a + 3\left(\frac{1}{3}\right) + 10\left(\frac{1}{5}\right) = 4 \tag{1.2.40.23}$$
$$2a + 1 + 2 = 4 \tag{1.2.40.24}$$
$$a = \frac{1}{2} \tag{1.2.40.25}$$

Since

$$a = \frac{1}{x}, \qquad b = \frac{1}{y}, \qquad c = \frac{1}{z} \tag{1.2.40.26}$$

Therefore,

$$x = 2 \tag{1.2.40.27}$$

$$y = 3 \tag{1.2.40.28}$$

$$z = 5 \tag{1.2.40.29}$$

1.2.41 If

$$A = \begin{pmatrix} 1 & -\cot x \\ \cot x & 1 \end{pmatrix}, \tag{1.2.41.1}$$

show that

$$A' A^{-1} = \begin{pmatrix} -\cos 2x & \sin 2x \\ -\sin 2x & -\cos 2x \end{pmatrix} \tag{1.2.41.2}$$

where A' denotes the transpose of A.
Solution: : Given,

$$A = \begin{pmatrix} 1 & -\cot x \\ \cot x & 1 \end{pmatrix} \tag{1.2.41.3}$$

Step 1: Find the transpose of A.
Interchanging the rows and columns, we get

$$A' = \begin{pmatrix} 1 & \cot x \\ -\cot x & 1 \end{pmatrix} \tag{1.2.41.4}$$

Step 2: Find the inverse of A.
For a matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \tag{1.2.41.5}$$

the inverse is

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \tag{1.2.41.6}$$

The determinant of A is

$$\left|A\right| = (1)(1) - (-\cot x)(\cot x). \tag{1.2.41.7}$$

Therefore,

$$\left|A\right| = 1 + \cot^2 x = \csc^2 x. \tag{1.2.41.8}$$

Hence,

$$A^{-1} = \frac{1}{1 + \cot^2 x} \begin{pmatrix} 1 & \cot x \\ -\cot x & 1 \end{pmatrix} \tag{1.2.41.9}$$

Step 3: Calculate $A' A^{-1}$.

Multiplying the matrices,

$$A'A^{-1} = \frac{1}{1+\cot^2 x}\begin{pmatrix} 1 & \cot x \\ -\cot x & 1 \end{pmatrix}\begin{pmatrix} 1 & \cot x \\ -\cot x & 1 \end{pmatrix} \tag{1.2.41.10}$$

Performing the multiplication,

$$A'A^{-1} = \frac{1}{1+\cot^2 x}\begin{pmatrix} 1-\cot^2 x & 2\cot x \\ -2\cot x & 1-\cot^2 x \end{pmatrix} \tag{1.2.41.11}$$

Therefore,

$$A'A^{-1} = \begin{pmatrix} \frac{1-\cot^2 x}{1+\cot^2 x} & \frac{2\cot x}{1+\cot^2 x} \\ -\frac{2\cot x}{1+\cot^2 x} & \frac{1-\cot^2 x}{1+\cot^2 x} \end{pmatrix} \tag{1.2.41.12}$$

Step 4: Apply trigonometric identities. We know that

$$\frac{1-\cot^2 x}{1+\cot^2 x} = -\cos 2x \tag{1.2.41.13}$$

and

$$\frac{2\cot x}{1+\cot^2 x} = \sin 2x. \tag{1.2.41.14}$$

Substituting these identities,

$$A'A^{-1} = \begin{pmatrix} -\cos 2x & \sin 2x \\ -\sin 2x & -\cos 2x \end{pmatrix} \tag{1.2.41.15}$$

Hence proved.

$$A'A^{-1} = \begin{pmatrix} -\cos 2x & \sin 2x \\ -\sin 2x & -\cos 2x \end{pmatrix} \tag{1.2.41.16}$$

1.2.42 If A_1 denotes the area of the region bounded by $y^2 = 4x, x = 1$ and x-axis in the first quadrant and A_2 denotes the area of region bounded by $y^2 = 4x, x = 4$, Find the ratio $A_1 : A_2$.

Solution: The equation of the parabola is

$$y^2 = 4x. \tag{1.2.42.1}$$

In the first quadrant,

$$y = 2\sqrt{x}. \tag{1.2.42.2}$$

From (1.1.6.1) The area under a curve $y = f(x)$ between $x = a$ and $x = b$ is

$$A = \int_a^b y\,dx. \tag{1.2.42.3}$$

Step 1: Calculate the area A_1.

The region A_1 lies between $x = 0$ and $x = 1$.

Therefore,

$$A_1 = \int_0^1 2\sqrt{x}\,dx. \tag{1.2.42.4}$$

Writing the integrand using fractional powers,

$$A_1 = 2\int_0^1 x^{1/2}\,dx. \tag{1.2.42.5}$$

Integrating,

$$A_1 = 2\left[\frac{x^{3/2}}{3/2}\right]_0^1. \tag{1.2.42.6}$$

Hence,

$$A_1 = \frac{4}{3}\left[x^{3/2}\right]_0^1. \tag{1.2.42.7}$$

Substituting the limits,

$$A_1 = \frac{4}{3}(1-0). \tag{1.2.42.8}$$

Therefore,

$$A_1 = \frac{4}{3} \text{ square units.} \tag{1.2.42.9}$$

Step 2: Calculate the area A_2.
The region A_2 lies between $x = 0$ and $x = 4$.
Therefore,

$$A_2 = \int_0^4 2\sqrt{x}\,dx. \tag{1.2.42.10}$$

Integrating,

$$A_2 = 2\left[\frac{x^{3/2}}{3/2}\right]_0^4. \tag{1.2.42.11}$$

Thus,

$$A_2 = \frac{4}{3}\left[x^{3/2}\right]_0^4. \tag{1.2.42.12}$$

Substituting the limits,

$$A_2 = \frac{4}{3}\left(4^{3/2} - 0\right). \tag{1.2.42.13}$$

Since

$$4^{3/2} = \left(\sqrt{4}\right)^3 = 8, \tag{1.2.42.14}$$

we obtain

$$A_2 = \frac{4}{3} \times 8. \tag{1.2.42.15}$$

Therefore,

$$A_2 = \frac{32}{3} \text{ square units.} \tag{1.2.42.16}$$

Step 3: Find the ratio $A_1 : A_2$.
We have

$$A_1 : A_2 = \frac{4}{3} : \frac{32}{3}. \tag{1.2.42.17}$$

Multiplying both terms by 3,

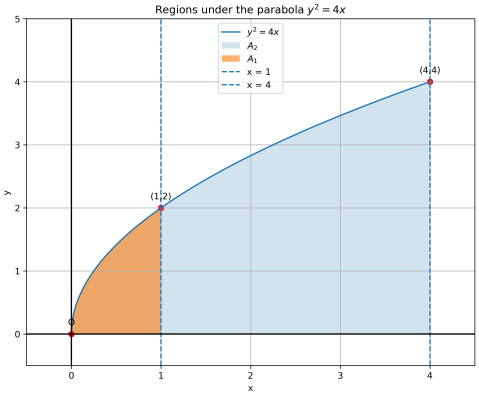
$$A_1 : A_2 = 4 : 32. \tag{1.2.42.18}$$

Dividing both terms by 4,

$$A_1 : A_2 = 1 : 8. \tag{1.2.42.19}$$

The following code gives the figure

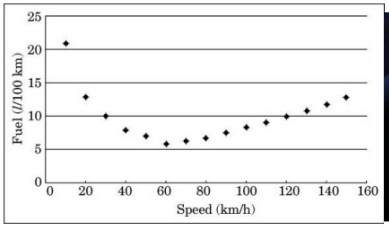
codes/s1225.py



1.2.43 Overspeeding increases fuel consumption and decreases fuel economy as a result of tyre rolling friction and air resistance. While vehicles reach optimal fuel economy at different speeds, fuel mileage usually decreases rapidly at speeds above 80km/h. The relation between fuel consumption F (litres per 100km) and speed V (km/h) is given by

$$F = \frac{V^2}{500} - \frac{V}{4} + 14. \tag{1.2.43.1}$$

On the basis of the above information,answer the following questions.





a) Find F when $V = 40$ km/h.
b) Find $\dfrac{dF}{dV}$.
c) (a) Find the speed V for which fuel consumption F is minimum.
d) (b) Find the quantity of fuel required to travel $600km$ at the speed V at which $\dfrac{dF}{dV} = -0.01$.

Solution: (i) Given,

$$F = \frac{V^2}{500} - \frac{V}{4} + 14. \tag{1.2.43.2}$$

Substituting $V = 40$,

$$F = \frac{40^2}{500} - \frac{40}{4} + 14. \tag{1.2.43.3}$$

Simplifying,

$$F = \frac{1600}{500} - 10 + 14. \tag{1.2.43.4}$$

Therefore,

$$F = 3.2 - 10 + 14. \tag{1.2.43.5}$$

Hence,

$$F = 7.2 \text{ litres per } 100 \text{ km}. \tag{1.2.43.6}$$

(ii) Given,

$$F = \frac{V^2}{500} - \frac{V}{4} + 14. \tag{1.2.43.7}$$

Differentiating both sides with respect to V,

$$\frac{dF}{dV} = \frac{1}{500}\frac{d}{dV}\left(V^2\right) - \frac{1}{4}\frac{d}{dV}\left(V\right) + \frac{d}{dV}\left(14\right). \tag{1.2.43.8}$$

Therefore,

$$\frac{dF}{dV} = \frac{2V}{500} - \frac{1}{4}. \tag{1.2.43.9}$$

Hence,

$$\frac{dF}{dV} = \frac{V}{250} - \frac{1}{4}. \tag{1.2.43.10}$$

(iii)
For minimum fuel consumption,

$$\frac{dF}{dV} = 0. \tag{1.2.43.11}$$

From (1.2.43.10)

$$\frac{V}{250} - \frac{1}{4} = 0. \tag{1.2.43.12}$$

Therefore,

$$\frac{V}{250} = \frac{1}{4}. \tag{1.2.43.13}$$

Multiplying both sides by 250,

$$V = \frac{250}{4}. \tag{1.2.43.14}$$

Hence,

$$V = 62.5 \text{ km/h}. \tag{1.2.43.15}$$

To verify that the fuel consumption is minimum, differentiate again:

$$\frac{d^2F}{dV^2} = \frac{1}{250}. \tag{1.2.43.16}$$

Since

$$\frac{d^2F}{dV^2} = \frac{1}{250} > 0, \tag{1.2.43.17}$$

the fuel consumption is minimum at $V = 62.5$ km/h.
Therefore, the required speed is

$$V = 62.5 \text{ km/h} \tag{1.2.43.18}$$

(iii) Find the quantity of fuel required to travel 600 km at the speed V for which

$$\frac{dF}{dV} = -0.01. \tag{1.2.43.19}$$

From (1.2.43.10)

$$\frac{dF}{dV} = \frac{V}{250} - \frac{1}{4}. \tag{1.2.43.20}$$

Given,

$$\frac{dF}{dV} = -0.01. \tag{1.2.43.21}$$

Therefore,

$$\frac{V}{250} - \frac{1}{4} = -0.01. \tag{1.2.43.22}$$

Rearranging,

$$\frac{V}{250} = \frac{1}{4} - 0.01. \tag{1.2.43.23}$$

Thus,

$$\frac{V}{250} = 0.25 - 0.01 = 0.24. \tag{1.2.43.24}$$

Multiplying both sides by 250,

$$V = 250\,(0.24)\,. \tag{1.2.43.25}$$

Hence,

$$V = 60 \text{ km/h}. \tag{1.2.43.26}$$

Step 2: Calculate fuel consumption at this speed.
The fuel consumption is

$$F = \frac{V^2}{500} - \frac{V}{4} + 14. \tag{1.2.43.27}$$

Substituting $V = 60$,

$$F = \frac{60^2}{500} - \frac{60}{4} + 14. \tag{1.2.43.28}$$

Simplifying,

$$F = \frac{3600}{500} - 15 + 14. \tag{1.2.43.29}$$

Therefore,

$$F = 7.2 - 15 + 14. \tag{1.2.43.30}$$

Hence,

$$F = 6.2 \text{ litres per } 100 \text{ km}. \tag{1.2.43.31}$$

Step 3: Calculate the fuel required for 600 km.
Since

$$\text{Fuel required for } 100 \text{ km} = 6.2 \text{ litres}, \tag{1.2.43.32}$$

the fuel required for 600 km is

$$\text{Fuel required} = \frac{600}{100} \times 6.2. \tag{1.2.43.33}$$

Therefore,

$$\text{Fuel required} = 6 \times 6.2. \tag{1.2.43.34}$$

Hence,

$$\text{Fuel required} = 37.2 \text{ litres}. \tag{1.2.43.35}$$

Final Answers:

$$\begin{aligned} &(i) && F = 7.2 \text{ litres per } 100 \text{ km}, \\ &(ii) && \frac{dF}{dV} = \frac{V}{250} - \frac{1}{4}, \\ &(iii)(a) && V = 62.5 \text{ km/h}, \\ &(iii)(b) && 37.2 \text{ litres for } 600 \text{ km}. \end{aligned} \tag{1.2.43.36}$$

1.2.44 The month of september is celebrated as the Rashtriya Poshan Maah across the country. Following a healthy and well-balanced diet is crucial in order to supply the body with the proper nutrients it needs. A balanced diet also keeps us mentally fit and promotes improved level of energy.

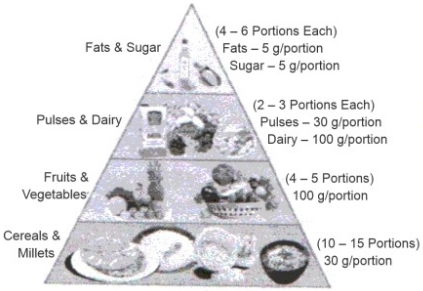


Figure-1

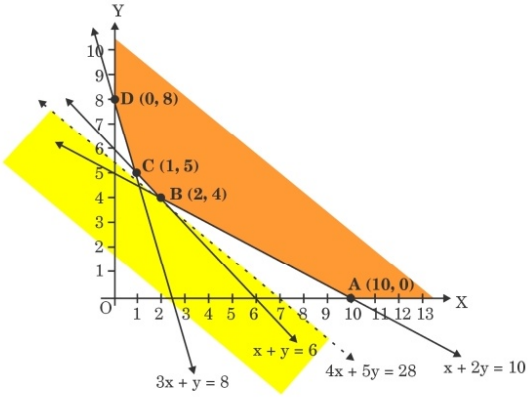


Figure-2

A dietician wishes to minimize the cost of a diet involving two types of food, food X (xkg) and food Y (ykg), which are available at the rates of Rs. 16/kg and Rs. 20/kg, respectively. The feasible region is shown in Figure 2.

Answer the following questions:

(i) Identify and write all the constraints which determine the given feasible region.

(ii) If the objective is to minimize the cost $Z = 16x + 20y$, find the values of x and y at which the cost is minimum. Also, find the minimum cost, assuming that a minimum exists for the given unbounded region.

Solution: i From the given graph, the corner points of the feasible region are

$$A\left(10,0\right),\quad B\left(2,4\right),\quad C\left(1,5\right),\quad D\left(0,8\right). \tag{1.2.44.1}$$

The boundary lines are

$$x+2y=10, \tag{1.2.44.2}$$
$$x+y=6, \tag{1.2.44.3}$$

and

$$3x+y=8. \tag{1.2.44.4}$$

From the shaded feasible region, the constraints are

$$x+2y=10, \tag{1.2.44.5}$$
$$x+y=6, \tag{1.2.44.6}$$
$$3x+y=8. \tag{1.2.44.7}$$

Since the quantities of food cannot be negative,

$$x=0,\qquad y=0. \tag{1.2.44.8}$$

Therefore, the required constraints are

$$x+2y=10, \tag{1.2.44.9}$$
$$x+y=6, \tag{1.2.44.10}$$
$$3x+y=8, \tag{1.2.44.11}$$
$$x=0, \tag{1.2.44.12}$$
$$y=0. \tag{1.2.44.13}$$

ii The objective function is

$$Z=16x+20y. \tag{1.2.44.14}$$

The corner points of the feasible region are

$$A\left(10,0\right),\quad B\left(2,4\right),\quad C\left(1,5\right),\quad D\left(0,8\right). \tag{1.2.44.15}$$

We calculate the value of Z at each corner point.
At point $A\left(10,0\right)$:

$$Z=16\left(10\right)+20\left(0\right) \tag{1.2.44.16}$$
$$Z=160 \tag{1.2.44.17}$$

At point $B\left(2,4\right)$:

$$Z=16\left(2\right)+20\left(4\right) \tag{1.2.44.18}$$
$$Z=32+80 \tag{1.2.44.19}$$
$$Z=112 \tag{1.2.44.20}$$

At point $C\,(1,5)$:

$$Z = 16\,(1) + 20\,(5) \tag{1.2.44.21}$$

$$Z = 16 + 100. \tag{1.2.44.22}$$

$$Z = 116 \tag{1.2.44.23}$$

At point $D\,(0,8)$:

$$Z = 16\,(0) + 20\,(8) \tag{1.2.44.24}$$

$$Z = 160 \tag{1.2.44.25}$$

The results can be summarised in the following table:

Corner point	$Z = 16x + 20y$
$A\,(10,0)$	160
$B\,(2,4)$	112
$C\,(1,5)$	116
$D\,(0,8)$	160

(1.2.44.26)

The smallest value is

$$Z_{\min} = 112 \tag{1.2.44.27}$$

at the point

$$B\,(2,4)\,. \tag{1.2.44.28}$$

Step 2: Verify the minimum for the unbounded region.
Since the feasible region is unbounded, we verify that a smaller value of Z is not possible.
Suppose

$$Z < 112. \tag{1.2.44.29}$$

Then

$$16x + 20y < 112. \tag{1.2.44.30}$$

Dividing by 4,

$$4x + 5y < 28. \tag{1.2.44.31}$$

However, every point in the feasible region satisfies

$$4x + 5y = 28. \tag{1.2.44.32}$$

Therefore, no feasible point can satisfy $Z < 112$.
Hence, $Z = 112$ is indeed the minimum cost.
The minimum cost is

$$Z_{\min} = \text{Rs. } 112. \tag{1.2.44.33}$$

1.2.45 Airplanes are by far the safest mode of transportation when the number of transported passengers are measured against personal injuries and fatality totals. Previous records state that the probability of an airplane crash is 0.00001%. Further, there are 95% chances that there will be survivors after a plane crash. Assume that in case of no crash, all passengers survive.





Let E_1 be the event that there is a plane crash, E_2 be the event that there is no crash. Let A be the event that passengers survive after the journey.
Solution: (i) Find the probability that the airplane will not crash.
Given,

$$P\left(E_1\right) = 0.00001\%. \tag{1.2.45.1}$$

Converting the percentage into a probability,

$$P\left(E_1\right) = \frac{0.00001}{100} = 10^{-7}. \tag{1.2.45.2}$$

Since E_2 is the complement of E_1,

$$P\left(E_2\right) = 1 - P\left(E_1\right). \tag{1.2.45.3}$$

Therefore,

$$P\left(E_2\right) = 1 - 10^{-7}. \tag{1.2.45.4}$$

Hence,

$$P\left(E_2\right) = 0.9999999 \tag{1.2.45.5}$$

(ii) Find $P\left(A \mid E_1\right) + P\left(A \mid E_2\right)$.
Solution: The probability of passengers surviving given that a crash occurs is

$$P\left(A \mid E_1\right) = 95\% = 0.95. \tag{1.2.45.6}$$

If there is no crash, all passengers survive. Therefore,

$$P\left(A \mid E_2\right) = 1. \tag{1.2.45.7}$$

Hence,

$$P\left(A \mid E_1\right) + P\left(A \mid E_2\right) = 0.95 + 1. \tag{1.2.45.8}$$

Thus,

$$P\left(A \mid E_1\right) + P\left(A \mid E_2\right) = 1.95 \tag{1.2.45.9}$$

$(iii)(a)$ Find $P\left(A\right)$.
Solution: Using the law of total probability,

$$P\left(A\right) = P\left(E_1\right) P\left(A \mid E_1\right) + P\left(E_2\right) P\left(A \mid E_2\right). \tag{1.2.45.10}$$

Substituting the known values,

$$P\left(A\right) = 10^{-7}\left(0.95\right) + \left(1 - 10^{-7}\right)\left(1\right). \tag{1.2.45.11}$$

Expanding,

$$P\left(A\right) = 0.95 \times 10^{-7} + 1 - 10^{-7}. \tag{1.2.45.12}$$

Therefore,

$$P\left(A\right) = 1 - 0.05 \times 10^{-7}. \tag{1.2.45.13}$$

Hence,

$$P\left(A\right) = 1 - 5 \times 10^{-9}. \tag{1.2.45.14}$$

$(iii)(b)$ Find $P\left(E_2 \mid A\right)$
Solution: Using Bayes' theorem ,

$$P\left(E_2 \mid A\right) = \frac{P\left(E_2\right) P\left(A \mid E_2\right)}{P\left(A\right)}. \tag{1.2.45.15}$$

From the previous parts,

$$P\left(E_2\right) = 1 - 10^{-7}, \qquad P\left(A \mid E_2\right) = 1, \tag{1.2.45.16}$$

and

$$P\left(A\right) = 1 - 5 \times 10^{-9}. \tag{1.2.45.17}$$

Substituting,

$$P\left(E_2 \mid A\right) = \frac{1 - 10^{-7}}{1 - 5 \times 10^{-9}}. \tag{1.2.45.18}$$

Therefore,

$$P\left(E_2 \mid A\right) = \frac{1 - 10^{-7}}{1 - 5 \times 10^{-9}} \tag{1.2.45.19}$$

C. Formulae-10th

1.3.1 Let $ax^2 + bx + c = 0$ be a quadratic equation.Let α, β be the roots of the equation.

$$\text{The roots of the equation are} \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{1.3.1.1}$$

$$\text{sum of the roots of the equation } = \alpha + \beta = -\frac{b}{a} \tag{1.3.1.2}$$

$$\text{product of the roots of the equation} = \alpha\beta = \frac{c}{a} \tag{1.3.1.3}$$

$$\text{Discriminant } = D = b^2 - 4ac. \tag{1.3.1.4}$$

$b^2 - 4ac = 0$, if roots are real and equal.
$b^2 - 4ac > 0$, if roots are real and distinct.
$b^2 - 4ac < 0$, if roots are complex.

1.3.2 Let $a, a_2, a_3, ..., a_n$ be the n terms of arithmetic progression.Then,

$$\text{common difference d} = a_n - a_{n-1} \tag{1.3.2.1}$$

$$\text{General equation for nth term of A.P series is } a_n = a + (n-1)\,d \tag{1.3.2.2}$$

$$\text{Sum of first n terms } S_n = \frac{n}{2}\left[2a + (n-1)\,d\right] \tag{1.3.2.3}$$

$$\text{Sum of first n terms } S_n = \frac{n}{2}\left[a + l\right] \tag{1.3.2.4}$$

1.3.3 Let $A\,(x_1, y_1)$ and $B\,(x_2, y_2)$ be two points then the distance between two points A and B is.

$$\mathbf{AB} = \mathbf{B} - \mathbf{A} = \begin{pmatrix} x_1 - y_1 \\ y_2 - y_1 \end{pmatrix} \tag{1.3.3.1}$$

$$\|AB\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \tag{1.3.3.2}$$

1.3.4 Let $x_1, x_2, ..., x_n$ be some data and $f_1, f_2, ..., f_n$ be their respective frequencies then the mean of the data is

$$\bar{x} = \frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i} \tag{1.3.4.1}$$

1.3.5 Surface area of sphere

$$S = 4\pi r^2 \tag{1.3.5.1}$$

1.3.6 Curved surface area of hemisphere

$$S = 2\pi r^2 \tag{1.3.6.1}$$

1.3.7 Total surface area of a hemisphere

$$S = 3\pi r^2 \tag{1.3.7.1}$$

1.3.8 Volume of the cone.

$$V = \frac{1}{3}\pi r^2 h \tag{1.3.8.1}$$

1.3.9 Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ then the determinant of matrix A is.

$$\left|A\right| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc \tag{1.3.9.1}$$

1.3.10 Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ then the inverse of matrix A exists only when $\left|A\right| \neq 0$ and the inverse is,

$$A^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \tag{1.3.10.1}$$

1.3.11 $sin\,(A + B) = sinAcosB + cosAsinB$

1.3.12 Let $P\,(x, y)$ divides the line segment joining $A\,(x_1, y_1)$ and $B\,(x_2, y_2)$ in the ratio m:n.

$$(x, y) = \left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n}\right) \tag{1.3.12.1}$$

1.3.13 Volume of the cylinder is

$$V = \pi r^2 h \tag{1.3.13.1}$$

1.3.14 Lenth of the arc of the circle subtending an angle θ is

$$l = \frac{\theta}{360^\circ} 2\pi r \tag{1.3.14.1}$$

1.3.15 Area of the arc of the circle subtending an angle θ is

$$l = \frac{\theta}{360^\circ} \pi r^2 \tag{1.3.15.1}$$

1.3.16 The formula for the median of grouped data is

$$Median = l + \left(\frac{\frac{N}{2} - cf}{f}\right) h \tag{1.3.16.1}$$

N= total frequency
cf = cummulative frequency of class preceding the median class
h = class size
l = lower limit of the median class

1.3.17 The formula for mode of grouped data is

$$Mode = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) h \tag{1.3.17.1}$$

f_1= frequency of the modal class
f_0 = frequency of class preceding the modal class

f_2 = frequency of the class succeeding the modal class
h = class size
l = lower limit of the modal class

D. Problems-10th

1.4.1 If the sum of the zeroes of the polynomial $P(x) = 2x^2 - k\sqrt{2}x + 1$is $\sqrt{2}$, then the value of k is:

a) $\sqrt{2}$ b) 2 c) $2\sqrt{2}$ d) $\frac{1}{2}$

Solution:
For the given polynomial,

$$P(x) = 2x^2 - k\sqrt{2}x + 1 \tag{1.4.1.1}$$

Comparing with $ax^2 + bx + c = 0$, we get

$$a = 2, \quad b = -k\sqrt{2}, \quad c = 1 \tag{1.4.1.2}$$

From (1.3.1.2),

$$\text{Sum of roots} = -\frac{b}{a} \tag{1.4.1.3}$$

Given that the sum of the roots is $\sqrt{2}$ and $a = 2$, $b = -k\sqrt{2}$,

$$\sqrt{2} = \frac{k\sqrt{2}}{2} \tag{1.4.1.4}$$

Hence, $k = 2$.

1.4.2 If the probability of a player winning a game is 0.79, then the probability of his losing the same game is :

a) 1.79 b) 0.31 c) 0.21% d) 0.21

Solution:
Let A be an event and P(A) be the probability of happening of event A. $P(A^c)$ be the probability of not happening of event A.

$$P(A) + P(A^c) = 1 \tag{1.4.2.1}$$

let A be the event of winning the game.
Given that P(A)=0.79
Then the probability of his losing the game is

$$P(A^c) = 1 - 0.79 \tag{1.4.2.2}$$

$$P(A^c) = 0.21 \tag{1.4.2.3}$$

1.4.3 If the roots of equation $ax^2 + bx + c = 0$,$a \neq 0$ are real and equal ,then which of the following relation is true ?

a) $a = \frac{b^2}{c}$ b) $b^2 = ac$ c) $ac = \frac{b^2}{4}$ d) $c = \frac{b^2}{a}$

Solution:
Given equation is $ax^2 + bx + c = 0$
Given that the roots of the above equation are real and equal.
Hence, Discriminant D = 0
From (1.3.1.4)

$$b^2 - 4ac = 0 \tag{1.4.3.1}$$

$$ac = \frac{b^2}{4} \tag{1.4.3.2}$$

Hence proved,

1.4.4 In an A.P.,if the first term a = 7,nth term a_n = 84 and the sum of first n terms $s_n = \frac{2093}{2}$, then n is equal to :

a) 22 b) 24 c) 23 d) 26

Solution:
Given that a = 7 , $a_n = 84$ and $s_n = \frac{2093}{2}$
From (1.3.2.4)

$$S_n = \frac{n}{2}\left[a_1 + l\right] \tag{1.4.4.1}$$

$$n = 23 \tag{1.4.4.2}$$

1.4.5 If two positive integers p and q can be expressed as $p = 18a^2b^4$ and $q = 20a^3b^2$, where a and b are prime numbers , then LCM (p, q) is :

a) $2a^2b^2$ b) $180a^2b^2$ c) $12a^2b^2$ d) $180a^3b^4$

Solution:
step 1:
Prime factorization of $18 = 2 \times 3^2$
Prime factorization of $20 = 2^2 \times 5$
To find the lowest common Multiple (LCM), take the higest power of each prime factor involved: $LCM\,(18, 20) = 2^2 \times 3^2 \times 5 = 4 \times 9 \times 5 = 180$
Step 2:
For the variable a: The terms are a^2 and a^3. The highest power is a^3.
For the variable b: The terms are b^4 and b^2 . The highest power is b^4.
step 3:
$LCM\,(p, q) = 180a^3b^4$

1.4.6 AD is a median of $\triangle ABC$ with vertices A$(5, -6)$,B$(6, 4)$ and C$(0, 0)$.Length AD is equal to :

a) $\sqrt{68}units$ b) $2\sqrt{15}units$ c) $\sqrt{101}units$ d) $10units$

Solution:
Since AD is a median, D is the midpoint of side BC.
Coordinates of D:

$$D = \left(\frac{6+0}{2}, \frac{4+0}{2}\right) = (3, 2) \tag{1.4.6.1}$$

$$\mathbf{A} = \begin{pmatrix} 5 \\ -6 \end{pmatrix} \text{and} \mathbf{D} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \tag{1.4.6.2}$$

The lenght AD is

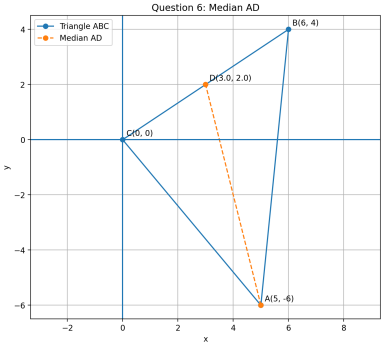
$$\mathbf{AD} = \mathbf{D} - \mathbf{A} = \begin{pmatrix} -2 \\ 8 \end{pmatrix} \tag{1.4.6.3}$$

$$\|AD\| = \sqrt{(-2)^2 + (8)^2} = \sqrt{68} \tag{1.4.6.4}$$

$$AD = \sqrt{68} \tag{1.4.6.5}$$

The following code generates the figure

codes/s6.py



1.4.7 If $sec\theta - tan\theta = m$, then the value of $sec\theta + tan\theta$ is :

a) $1 - \frac{1}{m}$ b) $m^2 - 1$ c) $\frac{1}{m}$ d) $-m$

Solution:
Given that

$$sec\theta - tan\theta = m \tag{1.4.7.1}$$

We know that

$$sec^2\theta - tan^2\theta = 1 \tag{1.4.7.2}$$

$$(sec\theta - tan\theta)(sec\theta + tan\theta) = 1 \tag{1.4.7.3}$$

From (1.4.7.1)

$$m(sec\theta + tan\theta) = 1 \tag{1.4.7.4}$$

$$sec\theta + tan\theta = \frac{1}{m} \tag{1.4.7.5}$$

1.4.8 From the data 1, 4, 7, 9, 16, 21, 25, if all the even numbers are removed , then the probability of getting at random a prime number from the remaining is :

a) $\frac{2}{5}$ b) $\frac{1}{5}$ c) $\frac{1}{7}$ d) $\frac{2}{7}$

Solution:

Given observations are 1, 4, 7, 9, 16, 21, 25.
If even numbers are removed then the remaining observations are 1, 7, 9, 21, 25 N(S) = 5.
Total no of observations that are in remaining data are 5.
Number of prime numbers in the remaining data is only 7 N(E) = 1.
Probability of getting a random prime number from remaining set is

$$P(E) = \frac{N(E)}{N(S)} \tag{1.4.8.1}$$

$$P(E) = \frac{1}{5} \tag{1.4.8.2}$$

1.4.9 For some data $x_1, x_2, \ldots, x_n$ with respective frequencies $f_1, f_2, \ldots, f_n$, the value of $\sum_{i=1}^{n} f_i(x_i - \bar{x})$ is equal to:

a) $n\bar{x}$ b) 1 c) Σf_i d) 0

Solution: The formula for the arithmetic mean($\bar{x}$) of a grouped frquency distribution is given by :

$$\bar{x} = \frac{\sum_{i=1}^{n} f_i x_i}{\sum_{i=1}^{n} f_i} \tag{1.4.9.1}$$

By cross-multiplying,we get the total sum of the data :

$$\sum_{i=1}^{n} f_i x_i = \bar{x} \sum_{i=1}^{n} f_i \tag{1.4.9.2}$$

Now, expand the given expression $\sum_{i=1}^{n} f_i (x_i - \bar{x})$:

$$\sum_{i=1}^{n} f_i (x_i - \bar{x}) = \sum_{i=1}^{n} f_i x_i - \sum_{i=1}^{n} f_i \bar{x} \tag{1.4.9.3}$$

Since $\bar{x}$ is a constant value (the mean),it can be taken outside the summation sign:

$$\sum_{i=1}^{n} f_i (x_i - \bar{x}) = \sum_{i=1}^{n} f_i x_i - \bar{x} \sum_{i=1}^{n} f_i \tag{1.4.9.4}$$

Substitute $\sum_{i=1}^{n} f_i x_i = \bar{x} \sum_{i=1}^{n} f_i$ into the equation:

$$\sum_{i=1}^{n} f_i (x_i - \bar{x}) = \sum_{i=1}^{n} f_i x_i - \bar{x} \sum_{i=1}^{n} f_i = 0 \tag{1.4.9.5}$$

Thus the sum of the deviations of a set of values from their mean is always zero.

1.4.10 The zeroes of a polynomial $x^2 + px + q$ are twice the zeroes of the polynomial $4x^2 - 5x - 6$. The value of p is :

a) $-\frac{5}{2}$ b) $\frac{5}{2}$ c) -5 d) 10

Solution:
Given quadratic equations are

$$x^2 + px + q = 0 \tag{1.4.10.1}$$

$$4x^2 - 5x - 6 = 0 \tag{1.4.10.2}$$

Given that zeroes of (1.4.10.1) are twice the zeroes of (1.4.10.2).
Sum of zeroes of equation (1.4.10.1) =$\frac{-p}{1}$
Sum of zeroes of equation (1.4.10.2) = $-\frac{-5}{4}$
Since given that zeroes of (1.4.10.1) are twice the zeroes of (1.4.10.2). So sum of

roots of equation (1.4.10.1) are twice of sum of equations of (1.4.10.2)
Therefore

$$\frac{-p}{1} = -2\frac{-5}{4} \tag{1.4.10.3}$$

$$p = -\frac{5}{2} \tag{1.4.10.4}$$

1.4.11 If the distance between the points $(3,-5)$ and $(x,-5)$ is 15 units , then the values of x are :

a) $12,-18$ b) $-12,18$ c) $18,5$ d) $-9,-12$

Solution:
Given two points are A$(3,-5)$ and B$(x,-5)$.
Given the distance between two points is 15 units.

$$\mathbf{AB} = \mathbf{B} - \mathbf{A} = \begin{pmatrix} x-3 \\ -5-(-5) \end{pmatrix} \tag{1.4.11.1}$$

From (1.3.3.2)

$$\|AB\| = \sqrt{(x-3)^2 + (-5-(-5))^2} = 15 \tag{1.4.11.2}$$

$$x - 3 = \pm 15 \tag{1.4.11.3}$$

From (1.4.11.3)

$$x - 3 = 15 \tag{1.4.11.4}$$

$$x = 18 \tag{1.4.11.5}$$

Also from (1.4.11.3)

$$x - 3 = -15 \tag{1.4.11.6}$$

$$x = -12,18 \tag{1.4.11.7}$$

1.4.12 If $cos\,(\alpha+\beta) = 0$, then value of $cos\left(\frac{\alpha+\beta}{2}\right)$ is equal to :

a) $\frac{1}{\sqrt{2}}$ b) $\frac{1}{2}$ c) 0 d) $\sqrt{2}$

Solution:

Given that

$$cos\left(\alpha+\beta\right)=0 \tag{1.4.12.1}$$

We know that

$$cos\left(\frac{\pi}{2}\right)=0 \tag{1.4.12.2}$$

Therefore from equations (1.4.12.1) and (1.4.12.2)

$$\alpha+\beta=\frac{\pi}{2} \tag{1.4.12.3}$$

$$\frac{\alpha+\beta}{2}=\frac{\pi}{4} \tag{1.4.12.4}$$

Therefore

$$cos\left(\frac{\alpha+\beta}{2}\right)=cos\left(\frac{\pi}{4}\right)=\frac{1}{\sqrt{2}} \tag{1.4.12.5}$$

1.4.13 A solid sphere is cut into two hemispheres. The ratio of the surface areas of sphere to that of two hemispheres taken together, is :

a) $1:1$ b) $1:4$ c) $2:3$ d) $3:2$

Solution:
Surface area of a sphere:
The total surface area of a solid sphere with radius r is:

$$\text{Surface Area of Sphere } = 4\pi r^2 \tag{1.4.13.1}$$

Surface Area of Two Hemispheres:
When a solid sphere is cut into two hemispheres, each hemisphere gets a new flat circular base.
The total surface area of one solid hemisphere = $3\pi r^2$.
Therefore, the total surface area of two hemispheres taken together is:

$$\text{Surface Area of Two Hemispheres } = 2\times 3\pi r^2 = 6\pi r^2 \tag{1.4.13.2}$$

Finding the ratio:

$$\frac{\text{Surface area of sphere}}{\text{Surface Area of Two Hemispheres}}=\frac{4\pi r^2}{6\pi r^2}=\frac{2}{3} \tag{1.4.13.3}$$

1.4.14 The volume of the largest right circular cone that can be carved out from a solid cube of edge 2 cm is :

a) $\frac{4\pi}{3}$ cu cm b) $\frac{5\pi}{3}$ cu cm c) $\frac{8\pi}{3}$ cu cm d) $\frac{2\pi}{3}$ cu cm

Solution:
For the cone to be the largest possible inside a cube of edge a = 2 cm :
The diameter of the cone's base equals the edge of the cube
$\rightarrow$ Diameter $= 2$ cm $\implies$ Radius $(r) = 1$ cm
The height of the cone equals the edge of the cube $\rightarrow$ Height $(h) = 2$ cm
The volume of a right circular cone is given by the formula:

$$V = \frac{1}{3}\pi r^2 h \tag{1.4.14.1}$$

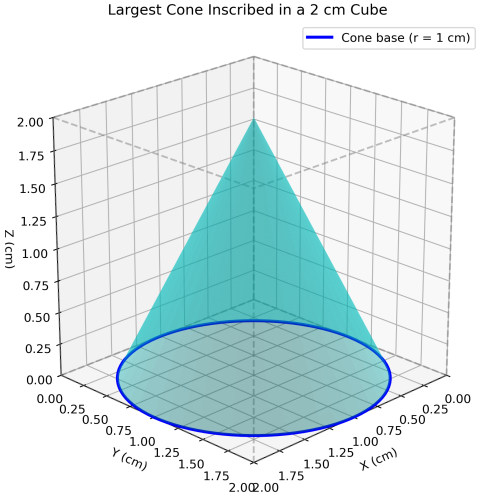
Substitute the values into the formula:

$$V = \frac{1}{3} \times \pi \times (1)^2 \times 2 = \frac{2\pi}{3} \text{ cu cm} \tag{1.4.14.2}$$

$$V = \frac{2\pi}{3} \text{ cu cm} \tag{1.4.14.3}$$

The following code generates the figure

codes/s14.py



1.4.15 Two dice are rolled together. The probability of getting sum of numbers on the two dice as 2, 3 or 5 is:

a) $\frac{7}{36}$ b) $\frac{11}{36}$ c) $\frac{5}{36}$ d) $\frac{4}{9}$

Solution:
Step 1: Find total outcomes When two six-sided dice are rolled, the total number of possible outcomes is:

$$\text{Total outcomes} = 6 \times 6 = 36 \tag{1.4.15.1}$$

Step 2: Find favorable outcomes Let's list the pairs that give a sum of 2, 3, or 5:

$$\text{Sum} = 2 \implies (1,1) \quad (1 \text{ outcome}) \tag{1.4.15.2}$$
$$\text{Sum} = 3 \implies (1,2), (2,1) \quad (2 \text{ outcomes}) \tag{1.4.15.3}$$
$$\text{Sum} = 5 \implies (1,4), (2,3), (3,2), (4,1) \quad (4 \text{ outcomes}) \tag{1.4.15.4}$$

Total favorable outcomes = 1 + 2 + 4 = 7
Probability of getting sum of numbers on the two dices as 2, 3 or 5 is

$$P(E) = \frac{\text{Total Favourable Outcomes}}{\text{Total Outcomes}} = \frac{7}{36} \tag{1.4.15.5}$$

1.4.16 The centre of a circle is at $(2,0)$. If one end of a diameter is at $(6,0)$, then the other end is at :

a) $(0,0)$ b) $(4,0)$ c) $(-2,0)$ d) $(-6,0)$

Solution:
Given centre of the circle is $(2,0)$.
Given one end of diameter is $(6,0)$.
Let the other end of diameter be (x,y).
Therefore,

$$(2,0) = \left(\frac{x+6}{2}, \frac{y+0}{2}\right) \tag{1.4.16.1}$$

$$(2,0) = \left(\frac{x+6}{2}, \frac{y}{2}\right) \tag{1.4.16.2}$$

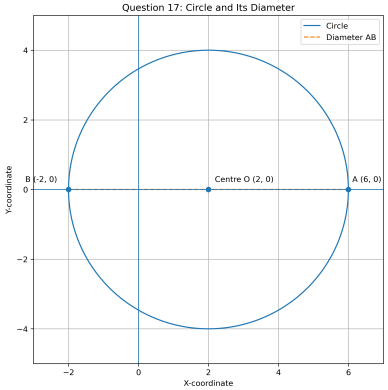
From (1.4.16) equating x coordinates on both sides

$$4 = x + 6 \tag{1.4.16.3}$$

$$x = -2 \tag{1.4.16.4}$$

From (17.1) y coordinates on both sides are equal.
Therefore other end of diameter of circle is $(-2,0)$.
The following code generates the figure

codes/s16.py



1.4.17 Solve the following system of linear equations 7x - 2y = 5 and 8x + 7y = 15 and verify your answer.

Solution: Given equations are

$$7x - 2y = 5 \tag{1.4.17.1}$$

$$8x + 7y = 15 \tag{1.4.17.2}$$

Writing the equations in matrix form, we get

$$AX = B \tag{1.4.17.3}$$

where

$$A = \begin{pmatrix} 7 & -2 \\ 8 & 7 \end{pmatrix}, \qquad X = \begin{pmatrix} x \\ y \end{pmatrix}, \qquad B = \begin{pmatrix} 5 \\ 15 \end{pmatrix} \tag{1.4.17.4}$$

Therefore,

$$X = A^{-1}B \tag{1.4.17.5}$$

First, find the determinant of matrix A.

$$\left|A\right| = (7)(7) - (-2)(8) \tag{1.4.17.6}$$

$$\left|A\right| = 49 + 16 = 65 \tag{1.4.17.7}$$

Since $\left|A\right| \neq 0$, the inverse of A exists from (1.3.10.1).

$$A^{-1} = \frac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \tag{1.4.17.8}$$

Hence,

$$A^{-1} = \frac{1}{65}\begin{pmatrix} 7 & 2 \\ -8 & 7 \end{pmatrix} \tag{1.4.17.9}$$

Now,

$$X = A^{-1}B \tag{1.4.17.10}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{65}\begin{pmatrix} 7 & 2 \\ -8 & 7 \end{pmatrix}\begin{pmatrix} 5 \\ 15 \end{pmatrix} \tag{1.4.17.11}$$

Multiplying the matrices, we get

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{65}\begin{pmatrix} 7\,(5) + 2\,(15) \\ -8\,(5) + 7\,(15) \end{pmatrix} \tag{1.4.17.12}$$

$$= \frac{1}{65}\begin{pmatrix} 65 \\ 65 \end{pmatrix} \tag{1.4.17.13}$$

Therefore,

$$x = 1, \qquad y = 1 \tag{1.4.17.14}$$

Verification:
Substituting $x = 1$ and $y = 1$ in (1.4.17.1),

$$7x - 2y = 7\,(1) - 2\,(1) = 7 - 2 = 5 \tag{1.4.17.15}$$

Hence, the first equation is verified. Substituting $x = 1$ and $y = 1$ in (1.4.17.2),

$$8x + 7y = 8\,(1) + 7\,(1) = 8 + 7 = 15 \tag{1.4.17.16}$$

Hence, the second equation is also verified.

$$x = 1, \quad y = 1 \tag{1.4.17.17}$$

1.4.18 In a pack of 52 playing cards, one card is lost. From the remaining cards, a card is drawn at random. Find the probability that the drawn card is the queen of hearts, if the lost card is a black card.

Solution:

Total number of cards in a pack = 52.
Since the lost card is black, the queen of hearts is still in the pack.
Therefore, the number of remaining cards is

$$52 - 1 = 51 \tag{1.4.18.1}$$

The number of queen of hearts cards remaining is 1.

Using the probability formula,

$$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} \tag{1.4.18.2}$$

Hence, the required probability is

$$P(\text{Queen of Hearts}) = \frac{1}{51} \tag{1.4.18.3}$$

1.4.19 Evaluate:

$$2\sqrt{2}\cos 45^\circ \sin 30^\circ + 2\sqrt{3}\cos 30^\circ \tag{1.4.19.1}$$

Solution:
We know that

$$\cos 45^\circ = \frac{1}{\sqrt{2}}, \qquad \sin 30^\circ = \frac{1}{2} \tag{1.4.19.2}$$

and

$$\cos 30^\circ = \frac{\sqrt{3}}{2} \tag{1.4.19.3}$$

Substituting these values, we get

$$= 2\sqrt{2}\left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{2}\right) + 2\sqrt{3}\left(\frac{\sqrt{3}}{2}\right) \tag{1.4.19.4}$$

$$2\sqrt{2}\cos 45^\circ \sin 30^\circ + 2\sqrt{3}\cos 30^\circ = 4 \tag{1.4.19.5}$$

1.4.20 If $A = 60^\circ$ and $B = 30^\circ$, verify that

$$\sin(A+B) = \sin A \cos B + \cos A \sin B \tag{1.4.20.1}$$

Solution: Given that

$$A = 60^\circ, \qquad B = 30^\circ \tag{1.4.20.2}$$

Left-Hand Side (LHS):

$$\sin(A+B) = \sin(60^\circ + 30^\circ) = 1 \tag{1.4.20.3}$$

Right-Hand Side (RHS):

$$\sin A \cos B + \cos A \sin B \tag{1.4.20.4}$$

Substituting the values of A and B, we get

$$= \sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ \tag{1.4.20.5}$$

We know that

$$\sin 60^\circ = \frac{\sqrt{3}}{2}, \qquad \cos 30^\circ = \frac{\sqrt{3}}{2} \tag{1.4.20.6}$$

$$\cos 60^\circ = \frac{1}{2}, \qquad \sin 30^\circ = \frac{1}{2} \tag{1.4.20.7}$$

Therefore substituting the above values in (1.4.20.5) we get,

$$= \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} + \frac{1}{2} \times \frac{1}{2} = 1 \tag{1.4.20.8}$$

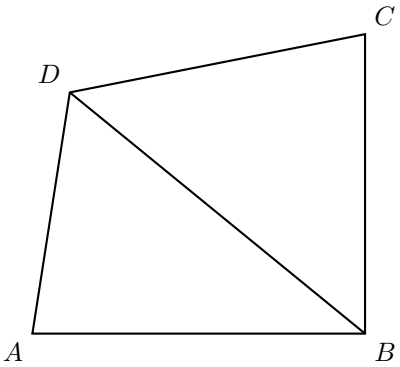
Thus,

$$\text{LHS} = \text{RHS} = 1 \tag{1.4.20.9}$$

$$\sin(A + B) = \sin A \cos B + \cos A \sin B \tag{1.4.20.10}$$

Hence, the identity is verified.

1.4.21 In the given figure, $ABCD$ is a quadrilateral. Diagonal BD bisects $\angle B$ and $\angle D$ both. Prove that:



(a) $\triangle ABD \sim \triangle CBD$

(b) $AB = BC$

Solution: (a) To prove: $\triangle ABD \sim \triangle CBD$

Since BD bisects $\angle B$,

$$\angle ABD = \angle CBD \tag{1.4.21.1}$$

Since BD bisects $\angle D$,

$$\angle ADB = \angle CDB \tag{1.4.21.2}$$

From equations (1) and (2), two angles of $\triangle ABD$ are equal to two angles of $\triangle CBD$ respectively.

Therefore, by the AA similarity criterion,

$$\triangle ABD \sim \triangle CBD \tag{1.4.21.3}$$

(b) To prove: $AB = BC$
Since,

$$\triangle ABD \sim \triangle CBD \tag{1.4.21.4}$$

sides are proportional. Hence,

$$\frac{AB}{CB} = \frac{BD}{BD} \tag{1.4.21.5}$$

Since BD is the common side,

$$\frac{AB}{CB} = 1 \tag{1.4.21.6}$$

Therefore,

$$AB = BC \tag{1.4.21.7}$$

Hence proved.

1.4.22 Prove that $5-2\sqrt{3}$ is an irrational number. It is given that $\sqrt{3}$ is an irrational number.
Solution: Let us assume that $5 - 2\sqrt{3}$ is rational.
Then,

$$5 - 2\sqrt{3} = r \tag{1.4.22.1}$$

where r is a rational number.
Rearranging,

$$2\sqrt{3} = 5 - r \tag{1.4.22.2}$$

$$\sqrt{3} = \frac{5 - r}{2} \tag{1.4.22.3}$$

Since 5, r, and 2 are rational numbers, $\frac{5 - r}{2}$ is also rational.
This contradicts the given fact that $\sqrt{3}$ is irrational.

$$5 - 2\sqrt{3} \text{ is irrational.} \tag{1.4.22.4}$$

$$\text{Hence proved.} \tag{1.4.22.5}$$

1.4.23 Show that the number $5 \times 11 \times 17 + 3 \times 11$ is a composite number.
Solution: Let

$$N = 5 \times 11 \times 17 + 3 \times 11 \tag{1.4.23.1}$$

$$N = 968 \tag{1.4.23.2}$$

Since $968 = 11 \times 88$, the number has factors other than 1 and itself.

$$968 \text{ is a composite number.} \tag{1.4.23.3}$$

1.4.24 Find the ratio in which the point $\left(\frac{8}{5}, y\right)$ divides the line segment joining the points $(1, 2)$ and $(2, 3)$. Also, find the value of y.

Solution:

Let $A\,(1, 2)$, $B\,(2, 3)$, and $P\left(\frac{8}{5}, y\right)$.

Let the ratio be $m : n$.

Using the section formula (1.3.12.1),

$$\frac{8}{5} = \frac{2m + 1n}{m + n} \tag{1.4.24.1}$$

$$\frac{m}{n} = \frac{3}{2} \tag{1.4.24.2}$$

Therefore, the required ratio is

$$3 : 2 \tag{1.4.24.3}$$

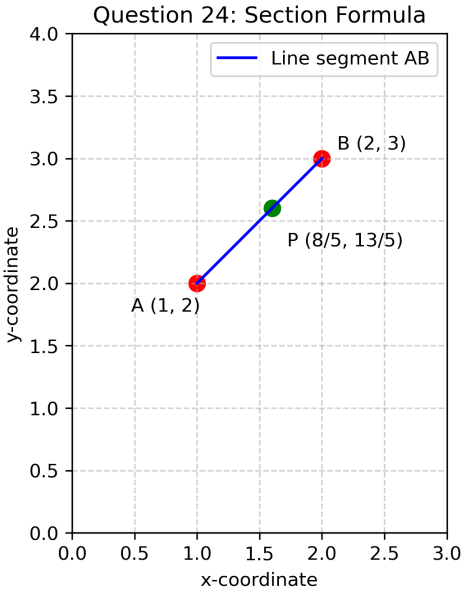
Now, using the section formula for the y-coordinate,

$$y = \frac{3\,(3) + 2\,(2)}{3 + 2} \tag{1.4.24.4}$$

$$y = \frac{13}{5} \tag{1.4.24.5}$$

The following code generates the figure

codes/s24.py



1.4.25 ABCD is a rectangle formed by the points $A(-1,-1)$, $B(-1,6)$, $C(3,6)$, and $D(3,-1)$. P, Q, R, and S are mid-points of sides AB, BC, CD, and DA respectively. Show that diagonals of the quadrilateral PQRS bisect each other.

Solution: Given:

$$A(-1,-1), \quad B(-1,6), \quad C(3,6), \quad D(3,-1) \tag{1.4.25.1}$$

Since P, Q, R, S are the mid-points of AB, BC, CD, DA, respectively, using the midpoint formula:

$$P = \left(\frac{-1+(-1)}{2}, \frac{-1+6}{2}\right) \tag{1.4.25.2}$$

similarly,

$$Q = (1,6), R = \left(3, \frac{5}{2}\right), S = (1,-1) \tag{1.4.25.3}$$

Now, the diagonals of quadrilateral $PQRS$ are PR and QS.
The midpoint of PR is

$$M_1 = \left(\frac{-1+3}{2}, \frac{\frac{5}{2}+\frac{5}{2}}{2}\right) \tag{1.4.25.4}$$

$$M_1 \left(1, \frac{5}{2}\right) \tag{1.4.25.5}$$

The midpoint of QS is

$$M_2 = \left(\frac{1+1}{2}, \frac{6+(-1)}{2}\right) \tag{1.4.25.6}$$

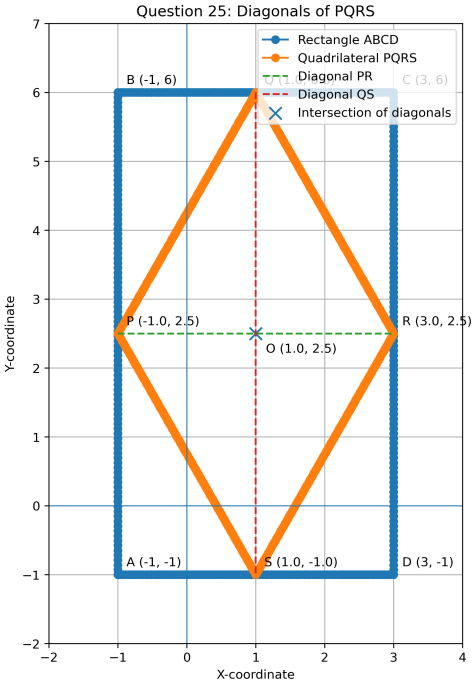
$$M_2 \left(1, \frac{5}{2}\right) \tag{1.4.25.7}$$

Since $M_1 = M_2$, the diagonals have the same midpoint.

The diagonals of $PQRS$ bisect each other. (1.4.25.8)

The following code generates the figure

codes/s25.py



1.4.26 In a workshop, 48, 80 and 144 teachers of three different subjects are present. If each room must contain the same number of teachers and each room must contain teachers of only one subject, find the minimum number of rooms required.

Solution: To find the minimum number of rooms, we calculate the HCF of 48, 80 and 144.

Prime factorisation:

$$48 = 2^4 \times 3 \tag{1.4.26.1}$$

$$144 = 2^4 \times 3^2 \tag{1.4.26.2}$$

Therefore,

$$\text{HCF}\,(48, 80, 144) = 2^4 = 16 \tag{1.4.26.3}$$

Hence, each room can contain 16 teachers.

Number of rooms required for the first subject:

$$\frac{48}{16} = 3 \tag{1.4.26.4}$$

Number of rooms required for the second subject:

$$\frac{80}{16} = 5 \tag{1.4.26.5}$$

Number of rooms required for the third subject:

$$\frac{144}{16} = 9 \tag{1.4.26.6}$$

Total number of rooms required:

$$3 + 5 + 9 = 17 \tag{1.4.26.7}$$

$$\text{Minimum number of rooms} = 17 \tag{1.4.26.8}$$

1.4.27 Prove that

$$\frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} = 1 + sec\theta cosec\theta \tag{1.4.27.1}$$

Solution: Taking the L.H.S.,

$$\begin{aligned}
\text{L.H.S.} &= \frac{\tan\theta}{1-\cot\theta} + \frac{\cot\theta}{1-\tan\theta} \\
&= \frac{\tan\theta\,(1-\tan\theta) + \cot\theta\,(1-\cot\theta)}{(1-\cot\theta)\,(1-\tan\theta)} \\
&= \frac{\tan\theta + \cot\theta - \tan^2\theta - \cot^2\theta}{1 - \tan\theta - \cot\theta + \tan\theta\cot\theta}
\end{aligned} \tag{1.4.27.2}$$

Using $\tan\theta\cot\theta = 1$,

$$\text{L.H.S.} = \frac{\tan\theta + \cot\theta - \tan^2\theta - \cot^2\theta}{2 - \tan\theta - \cot\theta} \tag{1.4.27.3}$$

Since

$$\tan^2\theta + \cot^2\theta = (\tan\theta + \cot\theta)^2 - 2, \tag{1.4.27.4}$$

we obtain

$$\text{L.H.S.} = \frac{\tan\theta + \cot\theta - (\tan\theta + \cot\theta)^2 + 2}{2 - \tan\theta - \cot\theta} \tag{1.4.27.5}$$

$$= \tan\theta + \cot\theta + 1 \tag{1.4.27.6}$$

Also,

$$\tan\theta + \cot\theta = \frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} \tag{1.4.27.7}$$

$$= \frac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} \tag{1.4.27.8}$$

Using

$$\sin^2\theta + \cos^2\theta = 1 \tag{1.4.27.9}$$

,

$$tan\theta + cot\theta = \frac{1}{\sin\theta cos\theta} = sec\theta cosec\theta \tag{1.4.27.10}$$

Therefore,

$$\text{L.H.S.} = 1 + sec\theta cosec\theta = \text{R.H.S.} \tag{1.4.27.11}$$

Hence proved.

1.4.28 Three years ago, Rashmi was thrice as old as Nazma. Ten years later, Rashmi will be twice as old as Nazma. How old are Rashmi and Nazma now?

Solution: Let the present age of Rashmi be x years and the present age of Nazma be y years.

Three years ago,

$$x - 3 = 3(y - 3) \tag{1.4.28.1}$$

$$x - 3y = -6 \tag{1.4.28.2}$$

Ten years later,

$$x + 10 = 2(y + 10) \tag{1.4.28.3}$$

$$x - 2y = 10 \tag{1.4.28.4}$$

Subtracting equation (1.4.28.2) from equation (1.4.28.4),

$$(x - 2y) - (x - 3y) = 10 - (-6) \tag{1.4.28.5}$$

$$y = 16 \tag{1.4.28.6}$$

Substituting $y = 16$ in equation (2),

$$x - 2\,(16) = 10 \tag{1.4.28.7}$$

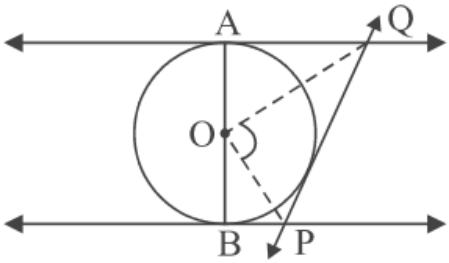
$$x = 42 \tag{1.4.28.8}$$

Therefore,

$$\text{Rashmi's present age} = 42 \text{ years} \tag{1.4.28.9}$$

$$\text{Nazma's present age} = 16 \text{ years} \tag{1.4.28.10}$$

1.4.29 In the given figure, AB is a diameter of the circle with centre O. AQ, BP and PQ are tangents to the circle. Prove that $\angle POQ = 90°$.



Solution: Since AQ and BP are tangents to the circle,

$$OA \perp AQ \tag{1.4.29.1}$$

$$OB \perp BP \tag{1.4.29.2}$$

Therefore,

$$\angle OAQ = 90°, \qquad \angle OBP = 90° \tag{1.4.29.3}$$

In quadrilateral $AOBP$,

$$\angle AOB = 180° \tag{1.4.29.4}$$

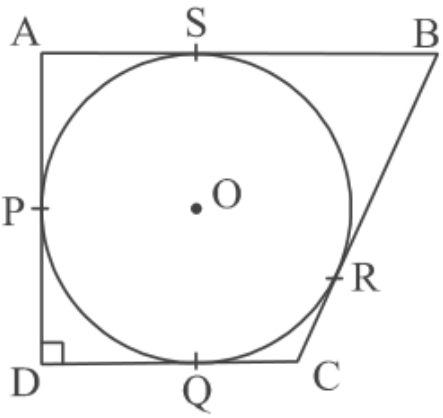
Since AB is a diameter, A, O, and B are collinear.
Using the properties of tangents drawn from an external point,

$$QA = \text{tangent from } Q, \qquad PB = \text{tangent from } P \tag{1.4.29.5}$$

Hence, the angle between PQ and PO and the angle between PQ and QO are complementary.
Therefore,

$$\angle POQ = 90^\circ \tag{1.4.29.6}$$

1.4.30 A circle with centre O and radius 8 cm is inscribed in a quadrilateral $ABCD$, in which P, Q, R, S are the points of contact as shown. If $AD \perp DC$, $BC = 30$ cm and $BS = 24$ cm, find the length DC.



Solution: Tangents drawn from an external point to a circle are equal.
From point B,

$$BS = BR = 24 \text{ cm} \tag{1.4.30.1}$$

Given,

$$BC = 30 \text{ cm} \tag{1.4.30.2}$$

Therefore,

$$BC = BR + RC \tag{1.4.30.3}$$

$$RC = 6 \text{ cm} \tag{1.4.30.4}$$

From point C,

$$CQ = CR = 6 \text{ cm} \tag{1.4.30.5}$$

Since $AD \perp DC$, and the circle has radius 8 cm, the tangent lengths from D to the points of contact on AD and DC are equal. Thus,

$$DQ = DP = 8 \text{ cm} \tag{1.4.30.6}$$

Therefore,

$$DC = DQ + QC \tag{1.4.30.7}$$

$$DC = 14 \text{ cm} \tag{1.4.30.8}$$

1.4.31 The difference between the outer and inner radii of a hollow right circular cylinder of length 14 cm is 1 cm. If the volume of the metal used in making the cylinder is 176 cm^3, find the outer and inner radii of the cylinder.

Solution: Let the outer radius be R cm and the inner radius be r cm. Given,

$$R - r = 1 \tag{1.4.31.1}$$

Therefore,

$$R = r + 1 \tag{1}$$

The volume of metal used in making the hollow cylinder is

$$V = \pi h \left(R^2 - r^2\right) \tag{1.4.31.2}$$

Given $h = 14$ cm and $V = 176$ cm^3,

$$\pi \times 14 \left(R^2 - r^2\right) = 176 \tag{1.4.31.3}$$

Using $\pi = \frac{22}{7}$,

$$\frac{22}{7} \times 14 \left(R^2 - r^2\right) = 176 \tag{1.4.31.4}$$

$$R^2 - r^2 = 4 \tag{1.4.31.5}$$

Using the identity $a^2 - b^2 = (a - b)(a + b)$,

$$(R - r)(R + r) = 4 \tag{1.4.31.6}$$

Since $R - r = 1$,

$$R + r = 4 \tag{2}$$

Adding equations (1) and (2),

$$2R = 5 \tag{1.4.31.7}$$

$$R = \frac{5}{2} = 2.5 \text{ cm} \tag{1.4.31.8}$$

Substituting $R = 2.5$ into $R - r = 1$,

$$2.5 - r = 1 \tag{1.4.31.9}$$

$$r = 1.5 \text{ cm} \tag{1.4.31.10}$$

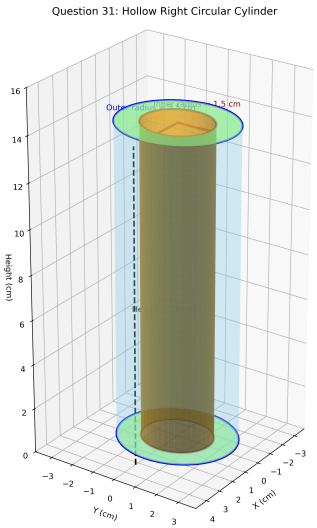
Therefore,

$$\text{Outer radius} = 2.5 \text{ cm} \tag{1.4.31.11}$$

$$\text{Inner radius} = 1.5 \text{ cm} \tag{1.4.31.12}$$

The following code generates the figure

codes/s31.py



1.4.32 An arc of a circle of radius 21 cm subtends an angle of 60° at the centre. Find:

(a) The length of the arc.

(b) The area of the minor segment of the circle made by the corresponding chord.

Solution: Given,

$$r = 21 \text{ cm}, \qquad \theta = 60^\circ \tag{1.4.32.1}$$

(a) Length of the arc

From (1.3.14.1) the length of an arc is given by

$$l = \frac{\theta}{360^\circ} \times 2\pi r \tag{1.4.32.2}$$

Substituting the values,

$$l = \frac{60^\circ}{360^\circ} \times 2 \times \frac{22}{7} \times 21 \tag{1.4.32.3}$$

$$l = 22 \text{ cm} \tag{1.4.32.4}$$

(b) Area of the minor segment

From (1.3.15.1) area of the minor segment

$$= \text{Area of sector } OAB - \text{Area of } \triangle OAB \tag{1.4.32.5}$$

Area of the sector is

$$= \frac{\theta}{360^\circ} \times \pi r^2 \tag{1.4.32.6}$$

$$= 231 \text{ cm}^2 \tag{1.4.32.7}$$

Since $OA = OB = 21$ cm and $\angle AOB = 60^\circ$, triangle OAB is equilateral. Therefore, its area is

$$= \frac{\sqrt{3}}{4} \times 21^2 \tag{1.4.32.8}$$

$$= \frac{441\sqrt{3}}{4} \text{ cm}^2 \tag{1.4.32.9}$$

Using $\sqrt{3} = 1.73$,

$$= \frac{441 \times 1.73}{4} \tag{1.4.32.10}$$

$$= 190.73 \text{ cm}^2 \tag{1.4.32.11}$$

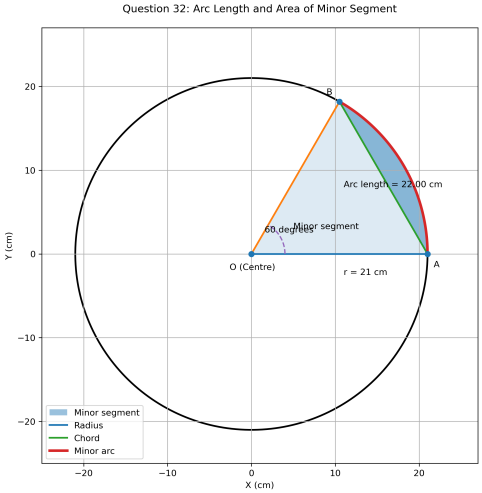
Hence, the area of the minor segment is

$$= 231 - 190.73 \tag{1.4.32.12}$$

$$\text{Area of minor segment} = 40.27 \text{ cm}^2 \tag{1.4.32.13}$$

The following code generates the figure

codes/s32.py



1.4.33 The sum of the first and eighth terms of an A.P. is 32 and their product is 60. Find the first term and common difference of the A.P. Hence, also find the sum of its first 20 terms.

Solution:

Let the first term be a and the common difference be d.
The first term is a, and the eighth term is $a + 7d$.
Given,

$$a + (a + 7d) = 32 \tag{1.4.33.1}$$

$$2a + 7d = 32 \tag{1.4.33.2}$$

Also, their product is 60.

$$a\,(a + 7d) = 60 \tag{1.4.33.3}$$

Let the first and eighth terms be x and y, respectively.
Then,

$$x + y = 32, \qquad xy = 60 \tag{1.4.33.4}$$

Therefore,

$$x\,(32 - x) = 60 \tag{1.4.33.5}$$

$$(x - 2)\,(x - 30) = 0 \tag{1.4.33.6}$$

Hence,

$$x = 2 \quad \text{or} \quad x = 30 \tag{1.4.33.7}$$

The corresponding pairs of first and eighth terms are $(2, 30)$ or $(30, 2)$.
Taking the first term as $a = 2$ and the eighth term as $a + 7d = 30$,

$$2 + 7d = 30 \tag{1.4.33.8}$$

$$a = 2, \quad d = 4 \tag{1.4.33.9}$$

From (**??**) the sum of the first 20 terms is

$$S_{20} = \frac{20}{2}\left[2a + (20 - 1)\, d\right] \tag{1.4.33.10}$$

Substituting $a = 2$ and $d = 4$,

$$S_{20} = 10\left[2\,(2) + 19\,(4)\right] \tag{1.4.33.11}$$

$$S_{20} = 800 \tag{1.4.33.12}$$

1.4.34 In an A.P. of 40 terms, the sum of the first 9 terms is 153 and the sum of the last 6 terms is 687. Determine the first term and common difference of the A.P. Also, find the sum of all the terms of the A.P.

Solution:

Let the first term be a and the common difference be d.
From (**??**) the sum of the first 9 terms is

$$S_9 = \frac{9}{2}\left[2a + (9 - 1)\, d\right] \tag{1.4.34.1}$$

Given $S_9 = 153$,

$$\frac{9}{2}\,(2a + 8d) = 153 \tag{1.4.34.2}$$

$$a + 4d = 17 \tag{1.4.34.3}$$

The last 6 terms are the 35th to 40th terms.
The 35th term is

$$a_{35} = a + 34d \tag{1.4.34.4}$$

The sum of the last 6 terms is

$$687 = \frac{6}{2}\left[2\,(a + 34d) + (6 - 1)\, d\right] \tag{1.4.34.5}$$

$$2a + 73d = 229 \tag{1.4.34.6}$$

Multiplying equation (1.4.34.3) by 2,

$$2a + 8d = 34 \tag{1.4.34.7}$$

Subtracting equation (1.4.34.7) from equation (1.4.34.6),

$$65d = 195 \tag{1.4.34.8}$$

$$d = 3 \tag{1.4.34.9}$$

Substituting $d = 3$ in equation (1.4.34.3),

$$a + 4\,(3) = 17 \tag{1.4.34.10}$$

$$a = 5 \tag{1.4.34.11}$$

Now, the sum of all 40 terms is

$$S_{40} = \frac{40}{2}\left[2a + (40 - 1)\,d\right] \tag{1.4.34.12}$$

$$S_{40} = 2540 \tag{1.4.34.13}$$

1.4.35 If a line is drawn parallel to one side of a triangle to intersect the other two sides at distinct points, prove that the other two sides are divided in the same ratio.

Solution: Given: In $\triangle ABC$, let D and E be points on AB and AC, respectively, such that $DE \parallel BC$.

To prove:

$$\frac{AD}{DB} = \frac{AE}{EC} \tag{1.4.35.1}$$

Since $DE \parallel BC$,

$$\triangle ADE \sim \triangle ABC \tag{1.4.35.2}$$

Therefore,

$$\frac{AD}{AB} = \frac{AE}{AC} \tag{1.4.35.3}$$

Since $AB = AD + DB$ and $AC = AE + EC$,

$$\frac{AD}{AD + DB} = \frac{AE}{AE + EC} \tag{1.4.35.4}$$

Cross-multiplying,

$$AD\,(AE + EC) = AE\,(AD + DB) \tag{1.4.35.5}$$

$$AD \cdot AE + AD \cdot EC = AD \cdot AE + AE \cdot DB \tag{1.4.35.6}$$

Cancelling $AD \cdot AE$ from both sides,

$$AD \cdot EC = AE \cdot DB \tag{1.4.35.7}$$

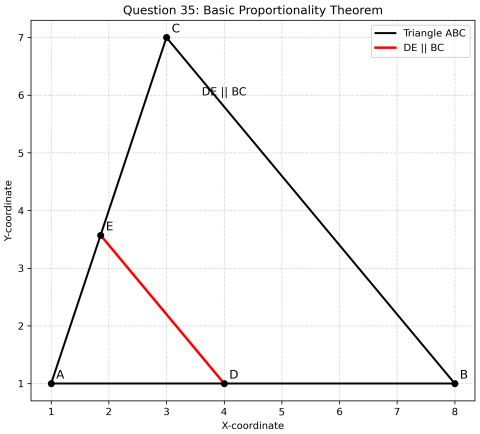
Hence,

$$\frac{AD}{DB} = \frac{AE}{EC} \tag{1.4.35.8}$$

Hence proved.

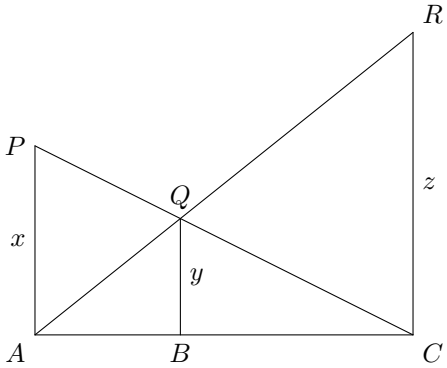
The following code generates the figure

codes/s35.py



1.4.36 In the given figure, PA, QB and RC are perpendicular to AC. If $AP = x$, $BQ = y$ and $RC = z$, prove that

$$\frac{1}{x} + \frac{1}{z} = \frac{1}{y}. \tag{1.4.36.1}$$



Solution: From the figure, A, B, C lie on the same straight line, and P, Q, R lie as shown.
Since $PA \perp AC$, $QB \perp AC$, and $RC \perp AC$,

$$\angle PAQ = \angle CBQ = 90^\circ \tag{1.4.36.2}$$

Also, $\angle PQA = \angle QCB$.
Therefore,

$$\triangle PAQ \sim \triangle CBQ \tag{1.4.36.3}$$

Hence,

$$\frac{PA}{BC} = \frac{AQ}{CQ} \tag{1.4.36.4}$$

Similarly, using the similar triangles formed by the intersecting lines,

$$\frac{BQ}{AP} = \frac{CQ}{AR} \tag{1.4.36.5}$$

Substituting $AP = x$, $BQ = y$, and $RC = z$, we obtain

$$\frac{1}{x} + \frac{1}{z} = \frac{1}{y} \tag{1.4.36.6}$$

Hence proved.

1.4.37 A tower is 6 m high. A point P on the ground is at a distance of 4 m from the foot of the tower. The angle of elevation of the top of the tower from P is $60°$. Find the height of the tower and the distance of point P from the foot of the tower. Take $\sqrt{3} = 1.73$.

Solution: Let AB be the tower and P be the point on the ground.
Given,

$$AB = 6 \text{ m}, \qquad \angle APB = 60° \tag{1.4.37.1}$$

Using the tangent ratio,

$$\tan 60° = \frac{AB}{BP} \tag{1.4.37.2}$$

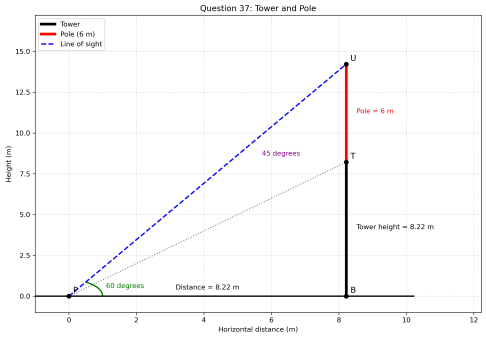
$$BP = 2\sqrt{3} \tag{1.4.37.3}$$

Using $\sqrt{3} = 1.73$,

$$BP = 3.46 \text{ m} \tag{1.4.37.4}$$

The following code generates the figure

codes/s37.py



1.4.38 A rectangular floor area can be completely tiled with 200 square tiles. If the side length of each tile is increased by 1 unit, it would take only 128 tiles to cover the floor.

(i) Assuming the original length of each side of a tile to be x units, make a quadratic equation from the above information.

(ii) Write the corresponding quadratic equation in standard form.

(iii) (a) Find the value of x, the length of the side of a tile, by factorisation.
OR
(b) Solve the quadratic equation for x, using the quadratic formula.



Solution:

(i) Forming the quadratic equation

Let the original side length of each square tile be x units.

Area of one original tile:x^2

Total floor area:$200x^2$

The increased side length of each tile is $(x+1)$ units.

Area of one increased tile:

$$(x+1)^2 \tag{1.4.38.1}$$

Since 128 increased tiles cover the same floor,

Therefore, the required quadratic equation is

$$200x^2 = 128\,(x+1)^2 \tag{1.4.38.2}$$

(ii) Standard form of the quadratic equation

Expanding,

$$200x^2 = 128\left(x^2+2x+1\right) \tag{1.4.38.3}$$

$$72x^2 - 256x - 128 = 0 \tag{1.4.38.4}$$

Dividing by 8,

$$9x^2 - 32x - 16 = 0 \tag{1.4.38.5}$$

(iii) (a) Solving by factorisation
The quadratic equation is

$$9x^2 - 32x - 16 = 0 \tag{1.4.38.6}$$

$$(9x + 4)(x - 4) = 0 \tag{1.4.38.7}$$

Therefore,

$$9x + 4 = 0 \quad \text{or} \quad x - 4 = 0 \tag{1.4.38.8}$$

$$x = -\frac{4}{9} \quad \text{or} \quad x = 4 \tag{1.4.38.9}$$

Since the side length cannot be negative,

$$x = 4 \text{ units} \tag{1.4.38.10}$$

OR
(iii) (b) Solving using the quadratic formula
The quadratic equation is

$$9x^2 - 32x - 16 = 0 \tag{1.4.38.11}$$

Comparing with $ax^2 + bx + c = 0$,

$$a = 9, \qquad b = -32, \qquad c = -16 \tag{1.4.38.12}$$

Using the quadratic formula,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{1.4.38.13}$$

Substituting the values,

$$x = \frac{32 \pm \sqrt{(-32)^2 - 4(9)(-16)}}{18} \tag{1.4.38.14}$$

$$x = \frac{32 \pm 40}{18} \tag{1.4.38.15}$$

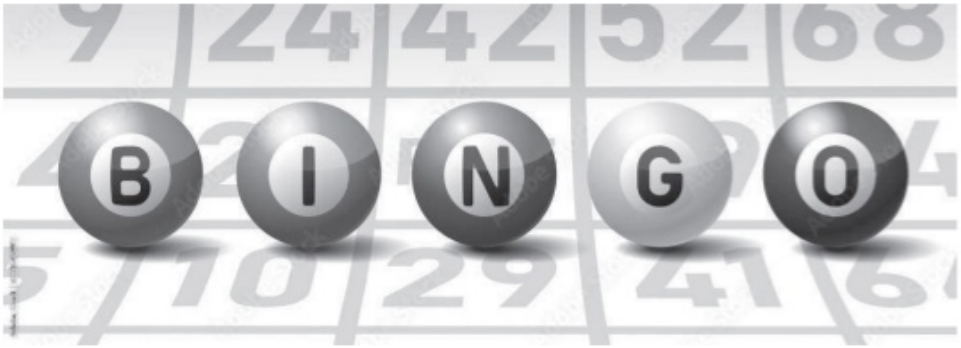
$$x = 4, -\frac{4}{9} \tag{1.4.38.16}$$

Rejecting the negative value,

$$x = 4 \text{ units} \tag{1.4.38.17}$$

1.4.39 BINGO is a game of chance. The host has 75 balls numbered 1 through 75. Each player has a BINGO card with some numbers written on it. The participant cancels

the number on the card when a number written on the ball selected at random is called out. Whoever cancels all the numbers on their card first says BINGO and wins the game. The following frequency distribution shows the numbers called out before



a player says BINGO:

Number on the ball	Frequency
$0-15$	8
$15-30$	9
$30-45$	10
$45-60$	12
$60-75$	9

Answer the following questions based on the above information:

(i) Write the median class.

(ii) What is the probability of calling an even number when the first ball is drawn?

(iii) (a) Find the median of the given data.

OR

(b) Find the mode of the given data.

Solution:

(i) Median class

The frequencies are

$$8,\ 9,\ 10,\ 12,\ 9 \tag{1.4.39.1}$$

The cumulative frequency table is:

Class interval	Frequency	Cumulative frequency
$0-15$	8	8
$15-30$	9	17
$30-45$	10	27
$45-60$	12	39
$60-75$	9	48

Total frequency:

$$N = 48 \tag{1.4.39.2}$$

$$\frac{N}{2} = \frac{48}{2} = 24 \tag{1.4.39.3}$$

The cumulative frequency immediately greater than 24 is 27, corresponding to the class $30 - 45$.
Therefore,

$$\text{Median class} = 30 - 45 \tag{1.4.39.4}$$

(ii) Probability of calling an even number
There are 75 balls numbered from 1 to 75.
The number of even numbers is 37.
Therefore,

$$P\left(\text{even number}\right) = \frac{\text{Number of even numbers}}{\text{Total number of balls}} \tag{1.4.39.5}$$

$$P\left(\text{even number}\right) = \frac{37}{75} \tag{1.4.39.6}$$

(iii) (a) Median of the given data
From (1.3.16.1) the formula for the median of grouped data is

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f}\right) h \tag{1.4.39.7}$$

Here,

$$l = 30, \quad N = 48, \quad cf = 17, \quad f = 10, \quad h = 15 \tag{1.4.39.8}$$

Substituting these values,

$$\text{Median} = 30 + \left(\frac{24 - 17}{10}\right) 15 \tag{1.4.39.9}$$

$$\text{Median} = 40.5 \tag{1.4.39.10}$$

OR
(iii) (b) Mode of the given data
The modal class is the class with the highest frequency.
The highest frequency is 12, so the modal class is $45 - 60$.
From (1.3.17.1) the formula for the mode of grouped data is

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) h \tag{1.4.39.11}$$

Here,

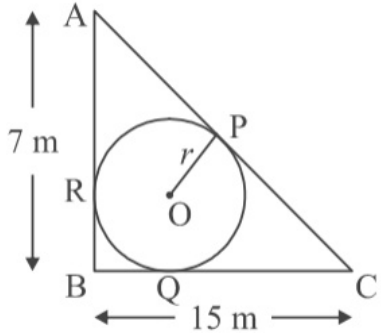
$$l = 45, \quad f_1 = 12, \quad f_0 = 10, \quad f_2 = 9, \quad h = 15 \tag{1.4.39.12}$$

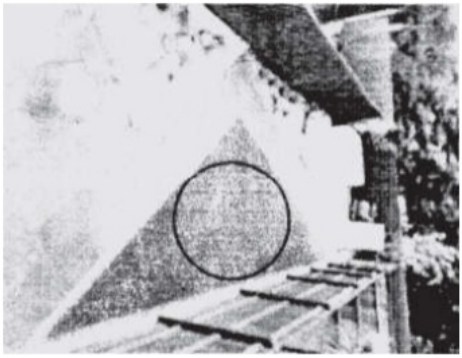
Substituting the values,

$$\text{Mode} = 45 + \left(\frac{12-10}{2\,(12)-10-9}\right)15 \tag{1.4.39.13}$$

$$\text{Mode} = 51 \tag{1.4.39.14}$$

1.4.40 A backyard is in the shape of a triangle ABC with a right angle at B. $AB = 7$ m and $BC = 15$ m. A circular pit was dug inside it such that it touches the walls AC, BC, and AB at P, Q, and R, respectively, such that $AP = x$ m. Based on the above





information, answer the following questions:

(i) Find the length of AR in terms of x.

(ii) Write the type of quadrilateral $BQOR$.

(iii) (a) Find the length PC in terms of x and hence find the value of x.

OR

(b) Find x and hence find the radius r of the circle.

Solution:

(i) Find the length of AR in terms of x

Tangents drawn from an external point to a circle are equal.

From point A,

$$AP = AR \tag{1.4.40.1}$$

Given $AP = x$ m,

$$AR = x \text{ m} \tag{1.4.40.2}$$

(ii) Write the type of quadrilateral $BQOR$

Since the radius is perpendicular to the tangent at the point of contact,

$$\angle BQO = 90^\circ \tag{1.4.40.3}$$

$$\angle BRO = 90^\circ \tag{1.4.40.4}$$

Also, $BQ = BR$ because tangents drawn from B are equal, and $OQ = OR$ because both are radii.
Therefore, $BQOR$ is a square.
(iii) (a) Find PC in terms of x and hence find x
Tangents drawn from point C are equal.
Therefore,

$$CP = CQ \tag{1.4.40.5}$$

From (1.4.40.2),

$$AB = AR + RB \tag{1.4.40.6}$$

$$RB = 7 - x \tag{1.4.40.7}$$

Since tangents drawn from B are equal,

$$BQ = BR = 7 - x \tag{1.4.40.8}$$

Now,

$$BC = BQ + QC \tag{1.4.40.9}$$

$$QC = 8 + x \tag{1.4.40.10}$$

Since $CP = CQ$,

$$PC = x + 8 \text{ m} \tag{1.4.40.11}$$

Using the Pythagorean theorem in right-angled triangle ABC,

$$AC^2 = AB^2 + BC^2 \tag{1.4.40.12}$$

$$AC = \sqrt{274} \text{ m} \tag{1.4.40.13}$$

Also,

$$AC = AP + PC \tag{1.4.40.14}$$

$$x = \frac{\sqrt{274} - 8}{2} \text{ m} \tag{1.4.40.15}$$

OR
(iii) (b) Find x and hence find the radius r of the circle
From the above result,

$$x = \frac{\sqrt{274} - 8}{2} \text{ m} \tag{1.4.40.16}$$

Since OQ is the radius and $BQOR$ is a square,

$$r = BQ \tag{1.4.40.17}$$

From (1.4.40.8), $BQ = 7 - x$
Therefore,

$$r = 7 - \frac{\sqrt{274} - 8}{2} \tag{1.4.40.18}$$

$$r = \frac{22 - \sqrt{274}}{2} \text{ m} \tag{1.4.40.19}$$